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Chapter 35: Endgame Conditions and Institutional Imagination—From Data Exhaustion to the Algorithmic Central Bank

Abstract

In 2025, the aggregate market capitalization of the key nodes hit by the Discrete Leviathan discriminant function (the several U.S. technology companies aggregated by the capital markets under the Mag 7 label) surpassed the total GDP of the EU-27. These nodes simultaneously control the compute layer, data layer, model layer, and distribution layer of the global digital infrastructure, and unilaterally set the internal parameters of AI models—the loss function, the Reinforcement Learning from Human Feedback (RLHF) reward, temperature—parameters that directly determine the quantity and quality of the cognitive output received daily by billions of people. No public institution has yet established an audit regime for the parameters of the loss function themselves.

This chapter first argues why the endgame conditions—data exhaustion, double performativity, and the attenuation of disciplinary memory—make "learning from mistakes" an institutionally unaffordable strategy in the AI domain. Then, proceeding from the institutional diagnosis of the fracture of the compensation loop and the analysis of Universal Basic Income (UBI) as circular rent, it argues that this institutional vacuum is not a matter of "the technology being too new and too complex to regulate in time"—but a structural absence, isomorphic in nature to the state of the financial domain before the establishment of the central bank system. Drawing on the evolutionary history of the central bank system in the financial domain, this chapter does not propose a perfect institutional design, but argues three things: why an Algorithmic Central Bank is needed, what it would govern,

and what the fundamental difficulties are.

Keywords

algorithmic central bank; loss function; data exhaustion; model collapse; compensation loop; public audit fund; cognitive debt

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§1 Data Exhaustion and Model Collapse — When the Mirror Reflects the Mirror

§1.1 The Geology of Data as a Resource

Previous chapters have described human data as the object of the Discrete Leviathan's primitive accumulation. Now we must ask: does this accumulation have boundaries?

Answering this question requires distinguishing between two kinds of data. *native data*—the externalized expressions such as text, images, and speech produced by humans under non-AI-mediated conditions—has its production rate constrained by biological limits: roughly 8 billion people, each with approximately 16 waking hours per day and an upper bound on the rate at which behavior externalizes information of approximately 10 bits per second (Zheng & Meister, 2025)—the same study stresses that sensory data-gathering rates are about eight orders of magnitude higher, so the constraint operates at the behavioral output end. Only the portion that humans enact and externalize into a recordable form precipitates as native data, and the total amount of information that can be externalized is first constrained by this behavioral throughput bound. *synthetic data*—data generated by AI for training the next generation of AI—has a production rate not constrained by biology. Using a geological metaphor: native data is like fossil fuels (energy compressed over eons, finite in total volume); synthetic data is like biofuel (theoretically renewable, but with far lower energy density, and its production itself consumes energy). Here "exhaustion" means that the expansion of the usable stock under given conditions is constrained, not that data is physically consumed in use; the 10-bits-per-second throughput bound limits the generation of behavioral information and does not translate directly into quantities of text, images, or training corpora. The "information density" of native data—its diversity, unexpectedness, and reflection of the real world's causal structures—is far higher than that of synthetic data, because the latter is a sample from the statistical distribution of the previous generation model; each sampling loses tail information and structural noise (and it is precisely this "noise" that contains the complexity of the real world).

Villalobos et al. (2022; revised 2024) projected that, if current trends continue, by the median crossing point in 2028, model training scales will approach the total effective stock of text in the indexed public web—around 4×10^{14} tokens; high-quality public human text will be fully utilized between 2026 and 2032 (median 2028). "Public human text" here is not identical with the "native data not mediated by AI" strictly defined in this chapter; the bound on the latter is the throughput argument of §1.1. Shumailov et al. (2024, *Nature*) precisely characterized the consequences of substituting synthetic data for native data: *model collapse*—in recursive training that indiscriminately pollutes the next generation's training set with model-generated outputs, models progressively lose information about the true underlying distribution, with the tails disappearing first, and learned behaviors converge over the generations to a point estimate with very small variance; the effect arises across learned generative models, and access to the original human data distribution proves crucial. Using the mirror metaphor: an infinite corridor of mirrors reflecting mirrors, each reflection loses resolution, until the image becomes an

unstructured blur. In the Macro-Cognitive Accounting Framework (MCAF), this corresponds to the threefold zeroing diagnosis of Chapter 28 §2: the starting point of verification is dismantled to zero already in single-generation models (the system performs no causal checking), while undifferentiated recursive training further degrades the informational depth of the output distribution—the loss of tails and the convergence of point estimates characterized by Shumailov can be read as the distributional form of this degradation.

Yu, Kim & Kim (2026, WWW '26) named the failure mode at the ecological layer *Retrieval Collapse*: in controlled simulations, when 67% of the corpus pool was AI-generated content, the search engine exposure contamination rate exceeded 80%, and the first phase presented an appearance of "seeming health"—the surface answer quality had not declined, but the diversity of information sources had already been systematically eroded. Spennemann (2025), by analyzing the frequency of ChatGPT characteristic words, estimated that at least 30% of active webpage text already comes from AI-generated sources, with the actual proportion possibly approaching 40%. This means: even if every AI model avoids training collapse by retaining some native data, the proportion of human signal in the entire public corpus pool shows a continuing downward trend. Once this degradation occurs—original sources overwritten by regenerated content with no independent archive—the lost distributional information cannot be recovered from the mixed signal; irreversibility depends on the retention of original material, not on the mixing process itself. This is not a crisis of model training; it is a crisis of the information ecology.

The Second Battlefield of Data Exhaustion. The expectation of approaching the data bottleneck and the competition for scarce domain data have already triggered systematic changes in acquisition patterns at the industry level. Zhang Yiping and Feng Shu (2026), using the U.S. Mercor platform as a sample, document in their analysis under anti-unfair competition law: as the dividends of open-source and public web data recede, the competition for scarce, proprietary domain data—non-public methods, internal analytical frameworks, and project experience from financial institutions, consulting firms, and law firms—has already shifted to the "second battlefield." Mercor acts as an intermediary between large AI labs and senior practitioners in specific industries, using baits of "expert consultation fees" of up to hundreds of dollars per hour to systematically recruit former employees from firms such as Goldman Sachs, Morgan Stanley, McKinsey, and Boston Consulting Group, inducing them, through the design of complex "challenges," to disclose non-public business information, technical information, and core knowledge meeting the constitutive requirements of trade secrets acquired during their prior careers—its essence is to decompose the data acquisition target "from the corporate entity into individual former employees," bypassing inter-enterprise compliance-sharing channels and pivoting to a "flanking attack."

This "external horizontal" acquisition channel and the Meta Model Capability Initiative (MCI) (the employer's longitudinal behavioral tracking of its own employees' work computers) already documented in Chapter 31, §4.3, constitute the structural comparison of the "second battlefield after data exhaustion"—internal vertical vs. external horizontal: when the expansion of usable public corpora under existing acquisition conditions approaches its bottleneck, the legitimacy boundaries of the next generation of training data are no longer determined by the internal dynamics of model collapse, but by where various acquisition channels that "bypass existing compliance classifications" are re-captured by the legal classification system. Chapter 31, §4.3 has already explicitly noted: the two channels share the same structural condition—existing legal classifications (labor contract / agency law / trade secret law / anti-unfair competition law) each cover only part of the picture, and the process of fully mapping them into damage categories for which legal remedies are available is still

taking shape in each jurisdiction.[1]

§1.2 Double Performativity: The Observer Changes the Observed

Model collapse and *double performativity* are two feedback loops that can couple: the former feeds model outputs directly into subsequent training; the latter has models alter human behavior, with the altered behavior then entering subsequent training. Together they widen the gap between training data and independent evidence from reality. Callon (1998) first proposed the "performativity" of economics—economic theory not only describes markets but participates in constructing them. MacKenzie (2006) concretized this insight as financial models not only describe markets—they shape markets: the widespread use of the Black-Scholes option pricing model caused actual prices to converge toward the model's predictions. This is single performativity: model → alters reality → reality converges toward model. MacKenzie's analysis was limited to the relationship between financial economics and financial markets.

AI introduces double performativity. The first layer: AI models alter user behavior (recommendation algorithms change reading habits, chatbots change writing styles). The second layer: the altered user behavior becomes the training data for the next generation of AI—what AI learns is no longer "the natural behavior of humans" but "human behavior that has been altered by the previous generation of AI." The coupling of the two layers of performativity produces a self-referential loop: AI shapes humans → the new behavior of humans trains AI → new AI further shapes humans → …

Double performativity causes "experience"—whether individual or disciplinary—to rapidly become obsolete. The "user behavior patterns" observed in 2020 had already been reshaped by successive rounds of recommendation-algorithm iteration by 2023. The world described by experience has already been changed by the very system that produces the experience—this is an amplified version of the observer effect at the macro-social level: the act of observation (AI training and deployment) systematically alters the observed (human behavioral data), making predictions based on historical observations fundamentally unreliable—the performativity described by MacKenzie has extended from financial models to the entire behavioral data ecology.

§1.3 The Ladder of Disciplinary Memory

Double performativity makes an already fragile condition even more fragile: disciplinary memory.

Information scientists Burton and Kebler proposed the concept of the literature "half-life" in 1960—borrowing the decay metaphor from radioactive substances to describe the rate at which the literature of a field becomes obsolete—and operationalized it using citation data (Burton & Kebler, 1960). Economist Fritz Machlup's contemporaneous economic analysis of knowledge production and knowledge obsolescence provided a broader

[1]Taking the regulatory sandbox—a typical instrument of gradualist regulatory learning—as the comparison layer: Guo Tengda and Zhou Daishu (2026) systematically reviewed the policy design of AI regulatory sandboxes in the U.S., UK, and EU for 2024–2025, summarizing them as "the scientificity and transparency of entry mechanisms / clear scope of exemptions / constructing a well-designed feedback closed-loop system / embedding risk control into each operational link." The core problem that sandboxes, as "exploratory governance tools," assume is that existing rules cannot precisely fit new technologies—but the structural changes revealed in §1.1 (exhaustion of public data → training data acquisition channels penetrating compliance classification boundaries) and the Algorithmic Central Bank (loss function / reward model / public boundary definition rights) proposed in §7 of this chapter operate at a level higher than what sandboxes can cover. This book cites Guo Tengda and Zhou Daishu (2026) solely as a "marker of the level of mainstream regulatory imagination" and does not also borrow their normative policy recommendations ("the tool attribute of balancing innovation promotion and regulatory learning / a multi-dimensionally differentiated entry assessment system / a policy closed loop that converts sandbox findings into institutional advantages," etc.).

background for this line of thought (Machlup, 1962). Samuel Arbesman later, in *The Half-Life of Facts*, synthesized a large number of quantitative studies, pointing out that the half-life of knowledge differs significantly across disciplines (Arbesman, 2012). This difference has direct implications for AI governance.

Zhao Dingxin, in his systematic analysis of the methodological differences between the natural sciences and the social sciences, provided a structural explanation for this difference (Zhao Dingxin, 2015). He distinguished three basic forms of knowledge—laws, covering laws, and mechanisms (Zhao Dingxin, 2015; Zhao Dingxin, 2022). A **law** refers to causal relationships that hold under a wide range of conditions—Newton's Second Law "always holds" in the macro, low-speed world, and is therefore a law. A **covering law** does not directly derive subordinate mechanisms but constrains the direction of mechanisms—the theory of evolution in biology belongs to this category: biological mechanisms are too numerous to enumerate, but "biological mechanisms that do not conform to evolutionary principles are unlikely to persist for long." A **mechanism** holds only under very limited and specific conditions—the law of price is a mechanism because it requires multiple conditions that are difficult to satisfy simultaneously in reality, such as perfect rationality, perfectly fluid information, and zero transaction costs. "Laws (principles) are mechanisms with broad applicability, and mechanisms are 'laws' that only hold under ideal conditions" (Zhao Dingxin, 2015).

This distinction directly explains the ladder of knowledge half-life: knowledge built on laws is naturally long-lived, because the conditions for the law's validity are satisfied almost everywhere and will not change because humans forget it; knowledge built on mechanisms is naturally fragile, because the conditions for the mechanism's validity can—and often do—be altered by human action. Zhao Dingxin further pointed out that paradigm shifts in the natural sciences are **irreversible**—"once a new paradigm is established, the old paradigm and corresponding theories either lose their value… or their value will be clearly locked within a certain range"; whereas the developmental path of the social sciences is the "**alternation** of mainstream paradigms under multiple paradigms (paradigm alternation), not paradigm shift"—"you finish your act, and I take the stage" (Zhao Dingxin, 2015).

Physics lies at one end of the spectrum—it is built upon laws. Newton's three laws, proposed in 1687, have had their scope of applicability delimited by general relativity and quantum mechanics, but remain valid as approximations—they have not been "forgotten." Using Zhao Dingxin's terminology, the emergence of relativity "locked the application scope of Newtonian mechanics" without causing it to lose its value—this is the irreversibility of a paradigm shift. There is another, more direct reason for the longevity of physics' memory: apples fall by themselves. The empirical verification of physical laws does not depend on human institutional memory—even if all textbooks were burned, the next generation would still observe apples falling. In MCAF language, part of the legitimacy base ($\|L\|$) of physics is anchored in physical reality external to human institutions.

Finance lies at the other end of the spectrum—it is built upon mechanisms, not laws. The Financial Instability Hypothesis proposed by Minsky in his 1986 book *Stabilizing an Unstable Economy* described an endogenous dynamic: periods of stability breed excessive risk-taking, leverage accumulates, and ultimately self-destructs (Minsky, 1986). This dynamic was highly consistent with the unfolding of the 2008 subprime crisis—yet during the decade before the crisis, Minsky was almost forgotten in mainstream finance. After the crisis, he was "rediscovered," but by the mid-2020s, financial markets once again exhibited many of the features he described (rising leverage ratios, compressed risk premiums), as if the lesson had never occurred. This is precisely the pattern of paradigm alternation described by Zhao Dingxin: knowledge in finance is not irreversibly replaced

by better knowledge, but cycles between forgetting and rediscovery—"you finish your act, and I take the stage." The obsolescence of financial knowledge lacks the automatic correction of external reality that physical knowledge has—money does not remember by itself that it has been devalued. The legitimacy base ($\|L\|$) of finance depends on institutional memory (regulatory rules, risk models, institutional culture) for its maintenance, and institutional memory is eroded by personnel turnover, profit pressures, and narratives of "this time is different."

AI engineering may lie at the extreme end of the spectrum—it is not only built upon mechanisms, but the mechanisms themselves have extremely short lifespans. A 1966 report in *Spectrum*, a publication of the IEEE (Institute of Electrical and Electronics Engineers), entitled "Technical Obsolescence," estimated that by 1960, the half-life of knowledge for engineering degrees had already shortened to approximately 10 years (Petersen, 1966). At the 1978 ACM (Association for Computing Machinery) annual conference, Paul Armer of the Charles Babbage Institute, in his report "Obsolescence of the Computing Professional: The Problem and Possible Solutions," examined the then-current claim that the "half-life" of computing knowledge is only five years, and concluded that—though the evidence is indirect—a strong case can be made that the "half-life" of knowledge in data processing is much shorter than an individual's working lifetime (Armer, 1978).

The situation in AI engineering is more extreme than in general software engineering. BERT (Bidirectional Encoder Representations from Transformers), published in 2018, dominated NLP (Natural Language Processing) benchmarks from 2019–2021 and was largely superseded by the large language model paradigm by 2023—its period of dominance was less than five years (Devlin et al., 2019). An oral history by Quanta Magazine in 2025, based on interviews with 19 NLP researchers, recorded the firsthand experience of this transition—interviewee Julian Michael recalled that "so many projects were dropped on the floor when BERT was released"—and after the emergence of ChatGPT, the entire field was reshaped once again (Pavlus, 2025). Similarly, deep learning engineering experience considered best practice in 2015 (LSTM [Long Short-Term Memory], manual feature engineering) had largely become obsolete by 2020.

It must be noted: a systematic quantitative study of the half-life of AI engineering knowledge does not yet exist. The judgment of "five years" is an induction based on two adjacent paradigm transition windows, with a limited sample size. However, the direction is clear: since the ACM's discussion in 1978, the rate of knowledge obsolescence in the computing field has been continuously accelerating, and AI, as the subfield with the fastest iteration, almost certainly has a knowledge half-life shorter than the 10-year benchmark for engineering.

This extremely short experiential validity period has two causes. The first is the extreme speed of technological turnover itself—each new generation of models devalues the engineering experience of the previous generation. The second is the double performativity discussed in the previous subsection: the object described by experience—human behavioral data—has already been altered after the experience is generated. Understanding this through Zhao Dingxin's framework: physics is built on laws, and the conditions for the validity of laws (macro, low-speed) are not altered by human action; finance is built on mechanisms, and the conditions for the validity of mechanisms remain relatively stable over certain periods, allowing knowledge to accumulate through institutional memory; AI engineering is built on a more fragile structure—it does not deal with laws, nor even with mechanisms whose conditions are relatively stable, but with statistical distributions—and statistical distributions drift due to the deployment of the model itself. The legitimacy base ($\|L\|$) of AI engineering cannot be anchored in external reality as physics can, and finds it far harder to accumulate through institutional memory, as finance does—the speed of institutional construction is far slower than the speed of technological iteration.

The policy implication of this ladder is: **AI governance cannot rely on "learning from mistakes"**—because the lessons of mistakes may be washed away by the next round of technological iteration before they can be institutionalized. Financial regulation provides a reference: the cycle from the exposure of institutional deficiencies in Basel II in 2004, to crisis validation in 2008, to the Basel III remedy in 2010, involved a learning cycle of approximately six years. The paradigm transition from BERT to GPT-4 in the NLP field also took roughly five years, but the pace of AI's engineering iteration far exceeds the response capacity of regulatory institutions. In the AI domain, waiting for crises to provide lessons may be an institutionally unaffordable strategy.

A counterpoint can illuminate this: Guo Tengda and Zhou Daishu (2026) systematically reviewed the design of AI regulatory sandboxes in the U.S., the UK, and the EU—a progressive learning architecture that allows trial and error within controlled boundaries, accompanied by regulatory observation, and adjusts rules after errors occur. Sandboxes indeed enclose "learning from mistakes" within a fenced experimental field, but their underlying temporal sequence remains "release first, observe later, adjust later"—the calibration cycle is determined by the speed of the regulator's observation and review. Under the rapid iteration rhythm of AI engineering, measured by successive paradigm-shift windows, what sandboxes reduce is merely the externality damage of single trials, without addressing the core of the diagnosis in this section: the speed at which "lessons are digested by institutional memory" remains slower than the speed of "model generational turnover" that generates the lessons. Sandboxes are necessary mitigation tools but not sufficient epistemological alternatives—this point provides a counterpoint for the historical reference in §3 and the Algorithmic Central Bank argument in §7 below.

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§2 The Structure of the Institutional Vacuum

The preceding section completed two key arguments from the perspective of endgame conditions: the structural irreversibility of cognitive debt (§1.1–1.2) and the attenuation of disciplinary memory (§1.3). This section, building on that foundation, interrogates the specific structure of the institutional vacuum.

§2.1 The Valuation Scale and Infrastructure Concentration of Seven Companies

Chapter 33, §5 has already provided the basic scale data: the total market capitalization of the seven companies is approximately $20.8 trillion (as of early October 2025), exceeding the EU-27's GDP of roughly $19.4 trillion (2024 figures; see Chapter 33, §5). A structural observation is supplemented here: these companies are not merely "large"—they simultaneously control the digital infrastructure at four key layers:

- **The Compute Layer**: NVIDIA's GPU (Graphics Processing Unit) hardware monopoly coupled with CUDA (Compute Unified Device Architecture) ecosystem lock-in, plus the three major clouds (AWS Amazon Web Services, Azure Microsoft Cloud, GCP Google Cloud Platform). Virtually all cutting-edge AI training runs on this infrastructure.
- **The Data Layer**: Google Search covers roughly nine-tenths of the global search market, and Meta's platforms have over 3 billion monthly active users (public figures, circa 2025). Amazon's e-commerce data. Apple's device data. This covers virtually all scenarios of human digital behavior.
- **The Model Layer**: OpenAI (Microsoft-invested), Google DeepMind, Meta LLaMA, Anthropic (Amazon-invested). Virtually all global frontier large models come from this network.

- **The Distribution Layer**: The App Store, Google Play, and the recommendation algorithms of various platforms. These determine the channels through which content reaches end users.

Simultaneous control of all four layers is not a pluralistic landscape naturally formed by market competition. It is structural centralization—a small number of nodes simultaneously controlling the infrastructure, the data sources, the model development, and the content distribution.

§2.2 The Union of Three Powers

The current institutional vacuum can be described using a precise analogy: these companies simultaneously play the roles of legislator, central bank, and commercial bank.

- **"Legislator"**: They design the loss function—determining what the model "wants." The loss function is not a technical parameter—it is the "constitution" of the AI system, defining the optimization objective. This "constitution" is unilaterally formulated by the company's engineering team.
- **"Central Bank"**: They set inference parameters—temperature (which controls the randomness of the output: the higher the temperature, the more the model tends to sample from low-probability options, leading to more divergent output; the lower the temperature, the more the output concentrates on the highest-probability option), Top-p (nucleus sampling threshold, limiting the model to sample only from candidate words whose cumulative probability reaches p; the smaller p is, the more conservative the output), and sampling strategy—determining the quantity and quality of the "cognitive currency" issued with each output.
- **"Commercial Bank"**: They provide services directly to billions of users—"lending" cognitive output.

The union of three powers means there is no external check or balance. The financial domain took centuries to separate these three functions. In the AI domain, the three powers remain united.

§2.3 The Structural Isomorphism Between Platform Power and Central Bank Power

§2.1–2.2 have established the facts: a small number of entities simultaneously control the four layers of digital infrastructure and combine the three powers in one. This section argues why this power is, *in nature*, central-bank-level—and why the reasons it needs to be publicized are structurally isomorphic with the reasons the central bank was publicized.

§2.3.1 The Ontological Basis of the Isomorphism: Centralized Processing of Public Resources

The ontological basis of the isomorphism is not the rhetorical analogy that "both manage time," but a precise structural condition: **a small number of nodes centrally process public resources, and their processing behavior has systematic external effects and cannot be decentrally supplied**. The centralization of the central bank is "one currency, one interest rate per country"—a single interest rate decision affects the intertemporal allocation of all economic actors. The centralization of the platform is "one recommendation algorithm determines what one billion users see"—a single loss function shapes the cognitive environment of all users. For example, the Kuaishou short video recommendation system performs centralized sorting for hundreds of millions of daily active users, with optimization metrics being watch time, interaction rate, and 7-day retention (Lin et al., 2026, SIGIR). The scale of Douyin/TikTok is even larger. This is not a verbal similarity about "involving time"—this is a precise match of the structural condition of centrally processing public resources.

Bajaj et al. (2026) provide theoretical support from the perspective of multi-agent system safety. Their original core argument is: in interactive systems, what determines system behavior is not the properties of individual models, but the **interaction topology**—the structure of information flow; when the topology is centralized, increasing scale does not correct information cascades but *stabilizes* them. This book's application of this is: recommendation algorithms are precisely instances of this centralized topology—the information flow of a billion users is uniformly shaped by the gradient descent of a single loss function; individual user feedback cannot possibly reverse the system-level allocation logic. Bajaj et al. argue about topological safety among AI agents; we map the same structure onto the "platform-user" relationship: the platform is the sole high-efficiency node, users are passive receiving ends, and the topology dictates that power cannot be decentralized.

§2.3.2 A Comparison of Power Depth: Macro Time Price vs. Micro Attention Allocation

The central bank manages the **macro price** of time. The interest rate is "the price of time" (Chancellor, 2022)—adjusting the interest rate is adjusting the exchange ratio between "money now" and "money in the future." Through the interest rate, the central bank influences the intertemporal decisions of all economic actors: whether enterprises invest in long-cycle projects, whether individuals save rather than consume. But the central bank **cannot determine**: what you specifically spend your money on, what prices you see, what information you receive. The power of the central bank is indirect—it influences expectations through price signals, while individuals still retain the right to choose.

Algorithmic platforms manage the **micro allocation** of time. The recommendation algorithm directly determines whether a user dwells on a piece of content for 3 seconds or 30 minutes, and directly determines what users see; through repeated exposure and feedback structures, it systematically influences what users feel and what they are inclined to believe. Through real-time bidding systems, the algorithm prices attention by impression, click, and conversion events (CPM/CPC/CPA—billed per thousand impressions, per click, and per conversion action, respectively; see §2.3.3). The platform's power is direct—it bypasses the mediation of price signals and directly casts the user's cognitive environment.

This comparison is not the basis of the isomorphism argument (the basis of the isomorphism is in §2.3.1) but a reinforcing argument: if the central bank—which merely indirectly influences intertemporal allocation—needs to be publicized because it meets the three grounds for publicization (see §2.3.6), then platforms—which directly cast the cognitive environment with even greater penetration than the central bank—have even stronger grounds for publicization when they satisfy the same conditions.

§2.3.3 The Precise Pricing Mechanism of Attention: CPM/CPC/CPA

Algorithmic platforms not only "occupy time" but also price attention by impression, click, and conversion events through advertising bidding systems. Three main pricing mechanisms:

- **CPM (Cost Per Mille)**: The price an advertiser pays for 1,000 content impressions. Advertisers typically bid more for high-engagement traffic, and the CPM of high-engagement content accordingly trends higher—the algorithm therefore has a structural incentive to recommend the former.
- **CPC (Cost Per Click)**: The price an advertiser pays for 1 click. The click-through rate of clickbait and emotional content is usually much higher than that of serious content—the algorithm therefore has a structural incentive to recommend clickbait.

- **CPA (Cost Per Action)**: The price an advertiser pays for 1 conversion (purchase/registration). The conversion rate of impulse consumer goods is usually higher than that of long-term investment goods—the algorithm therefore has a structural incentive to recommend impulse consumption.

The crucial difference emerges here: the central bank's interest rate is a **single, publicly observable parameter** that acts on all economic actors and can be observed and anticipated by everyone. The "attention price" of algorithmic platforms consists of **billions of opaque parameters**—each piece of content, each user, each moment has a different price, and no one outside can audit it. This is not a "more complex interest rate"—this is **the pricing power itself being hidden**. The central bank's interest rate decisions require the publication of meeting minutes and hearings; the platform's attention pricing decisions are completed automatically within the gradient descent of the loss function, with no public accountability whatsoever.

§2.3.4 The Reshaping of the Loss Landscape and the Absence of Public Constraints

The central bank reshapes the **economic landscape** through the interest rate: an interest rate of 5% causes the net present value of long-cycle projects to plummet—these projects become "high-loss regions" within the investment landscape; an interest rate of 0% turns them into "low-loss regions." However, the central bank's objectives are public (price stability + full employment), with independence safeguards and accountability mechanisms.

Algorithmic platforms reshape the **cognitive landscape** through the loss function. A simplified recommender system loss function takes the form:

$$L = -\alpha \cdot \text{watch_time} - \beta \cdot \text{interaction} - \gamma \cdot \text{share} + \delta \cdot \text{diversity_penalty}$$

When the weight of the watch-time term, jointly with its range of variation, dominates the diversity term ($\alpha \cdot \Delta w \gg \delta \cdot \Delta d$), high-engagement content tends to become a "low-loss region" in the traffic landscape, and deep content tends to become a "high-loss region." Creators respond rationally: they produce high-engagement content—why do short video creators all place hooks in the first three seconds? Why do delivery riders risk their lives riding against traffic? Why do e-commerce sellers use nearly identical titles and cover images? Because the loss landscape defined by the algorithm makes these behaviors local optima. Here diversity_penalty denotes a penalty for insufficient diversity, and comparisons between weights must consider the scales and ranges of variation of the variables. This formula illustrates the formation of recommender incentives; the actual objective functions of delivery and e-commerce systems must be examined separately—the rider and seller cases illustrate structurally similar incentives, not the same function. Users passively consume: deep content gradually disappears from the visible subspace. **But the platform's objectives are private** (profit maximization), with no independence constraints and no public accountability.

Shu & Wei's (2026) **decision energy density** framework provides an analytical framework for this comparison. They define $E_i^d = \lambda_i \cdot \iota_i \cdot \rho_i$ (λ = decision generation rate, ι = impact magnitude, ρ = execution reach), arguing that when deployment friction decreases, the E^d of the most efficient node continuously rises, and if institutions lack boundary constraints, sovereignty shifts toward that node. They did not make the central bank analogy. This book substitutes E^d into both the central bank and the platform: the central bank's $\iota \times \rho$ comes from "one interest rate decision affecting the intertemporal allocation of all economic actors"; the platform's $\iota \times \rho$ comes from "one optimization update driven by the loss function altering the allocation of the daily content exposures—thousands per user—of a billion users." The decision energy densities of the two

types of institutions are compared here structurally only; the difference lies in λ: the platform's optimization decisions occur continuously at the millisecond scale; the central bank's interest rate decision occurs every six weeks—the platform's E^d is significantly higher than the central bank's. Comparable estimates of the decision energy densities of the two types of institutions have not yet been completed; this paragraph gives no numerical conclusion—while almost all of the constraints it faces fall on the output side, and constraints on the optimization objective itself (the fourth tier) are absent.

Zhou et al. (2026) support the same argument from another angle. Their original argument is: **objective selection failure** is a failure mode distinct from capability failure—scalar optimization is valid only under specific assumptions, and when heterogeneous objectives are compressed into a single numerical value, context-dependent constraints are systematically undermined. This book's application is: the platform's loss function is precisely the institutionalization of this scalar compression—compressing heterogeneous public objectives such as cognitive health, information diversity, and social trust into the single scalar of "engagement." This is not an engineering defect; it is an institutional defect.

Coyne (FAccT 2026) further argues that RLHF annotators play three normative roles (Extension, Evidence, Authority), and the design decisions of annotation pipelines and loss functions are therefore not merely technical decisions but also metaethical, moral, and political ones—this book summarizes the point as: the design of the loss function is a political choice. This book agrees with this judgment and pushes it to an institutional conclusion: since it is a political choice, it should not be made unilaterally by private entities.

§2.3.5 From Recommendation Platforms to Generative AI Platforms: The Intensification of Cognitive Substitution

Traditional algorithmic platforms (early Facebook, YouTube) primarily engaged in content recommendation; users still needed to produce their own content and make their own decisions. Generative AI platforms (ChatGPT, Claude, Copilot) are beginning to directly substitute for cognitive labor—writing, programming, analysis, decision-making.

Chalkidis & Søgaard (FAccT 2026) systematically review the empirical evidence on cognitive offloading: AI dependence is associated with a statistically significant decline in critical thinking, weakened brain connectivity, and the loss of the ability to perform skills independently. McGurk & Khachaturov (Cambridge, 2026) argue and predict: AI-saturated markets will form a "barbell economy"—the scarcity premium on which middle-tier cognitive labor relies for bargaining power will be persistently compressed, and the authors project its rents to approach zero. The workplace field-study data of Brynjolfsson et al. (2025) show that AI assistance raises the productivity of less-experienced workers far more than that of experienced ones (newcomers move down the experience curve faster), suggesting that the experience premium is being compressed.

This book's synthetic application of these three sets of findings is: the intensification of cognitive substitution extends the "centralized processing of public resources" of §2.3.1 from the content layer to the labor layer—platforms not only centrally process information distribution but are beginning to centrally process cognitive production itself. This upgrades the urgency of publicization from "protecting information diversity" to "protecting the skill stock of the labor force." The irreversibility of the latter lies in the fact that de-skilled laborers cannot be "printed out"—§8.3 will unfold the argument on this fundamental asymmetry.

§2.3.6 Three Grounds for the Publicization of the Central Bank

Why was the central bank publicized? Not because it "manages time"—but because the three grounds converge on the same conclusion:

Three Grounds for Subjecting Systemic Platform Power to Public Governance	Corresponding Platform Condition	Literature Support
① Money supply is a public good and cannot be decentrally supplied—a country can only have one base currency; it cannot be spontaneously coordinated by the market	The public attention pool and information ecology are public goods and cannot be decentrally supplied—network effects of search/recommendation have made information entry points highly concentrated; users cannot "choose" to exit the recommendation ecology	He et al. (2026): Large Language Model (LLM)-dominated websites in Common Crawl (a public web corpus) increased from 2.1% in 2022 to 29.4% in 2025; 2/3 of search queries contain at least one LLM-dominated website in the top 20 results—the information ecology is being homogenized in a direction not yet seen to reverse
② The interest rate has strong externalities—one bank's interest rate decision affects the balance sheets of all other banks and the investment decisions of the real economy	The output of the recommendation algorithm has strong externalities—one platform's content distribution strategy systematically affects users' cognitive health, information diversity, and social trust	Chalkidis & Søgaard (FAccT 2026): Externalities cascade from individual cognition to the stock of social human capital (see §2.3.5 for details)
③ Private monopoly of the right to issue currency generates systematic moral hazard (a historical diagnosis: the free banking and private note-issuing era; private mints had an incentive to inflate)	The *design right* over the loss function/mapping function is currently completely privatized—platforms have a structural incentive to compress multiple objectives into a single scalar (engagement/profit)	Zhou et al. (2026): Objective selection failure—scalar rewards compress heterogeneous objectives (see §2.3.4 for details)

The first two factual grounds (the non-substitutable public-good position and strong externalities) already hold for platforms; the third is a historical diagnosis rather than a factual ground—when the right of issuance is in fact privately monopolized, publicization is the customary remedy in the history of governance. The nature of the problem is therefore not limited to "market failure requiring policy subsidies"—the allocation of the right of issuance itself requires a public decision. The central bank does not "subsidize" the banking industry—it structurally separates the issuance of base money from private banks (deposit creation by commercial banks within public boundaries is addressed in §7.1.2). The "Algorithmic Central Bank" argued for in §7 does not "subsidize" AI safety—it argues why the design right over the loss function cannot be left entirely in private hands.

§2.3.7 Interest Rate → AI Form → Attention Price → Cognitive Friction Elimination

The argument of §2.3 can now form a causal chain with the core thesis of Chapter 34. Chapter 34 argued: the era of ultra-low rates (near-zero in 2008–2015 and 2020–2022, with a brief hiking cycle from late 2015 to mid-2019) lowered the capital constraint on long-horizon trial and error → multiple AI paths ran in parallel; rate hikes (2022–) restored "the price of time" → the fastest-monetizing paths gained a systematic survival advantage → LLMs won.

This chapter's §2.3 supplements the latter half of the causal chain: the survival pressure on LLMs → the need for rapid monetization → the need to maximize user engagement → the further reinforcement of the private structure of the loss function → CPM/CPC/CPA systematically biased toward high-engagement content (§2.3.3) → reshaping of the loss landscape (§2.3.4) → AI begins substituting for cognitive labor → eliminating cognitive friction (§2.3.5).

The complete causal chain: **Interest rates not only determine "what kind of AI is made" (Chapter 34) but also, through the form of AI, determine "how attention is priced" (§2.3.3–4), ultimately determining "which cognitive frictions are eliminated" (§2.3.5).** This is not a "side effect of technological progress"—this is interest rates, through the selection of AI forms, locking in "the elimination of cognitive friction" as a structural inevitability.

§2.3.8 The Fundamental Asymmetry: Skilled Laborers Cannot Be Printed Out

Financial authorities can "print money" and deploy public funds in a crisis—the Federal Reserve injecting liquidity (2008 liquidity facilities and quantitative easing; 2020 unlimited Quantitative Easing [QE]), the Treasury executing bailouts (2008 TARP [Troubled Asset Relief Program]). But when the de-skilling process described in §2.3.5 is complete, no monetary or fiscal policy can quickly "print out" a cohort of laborers with actual operational skills—the acquisition of skills requires time and cognitive friction; this is an insurmountable physical constraint.

This means the focus of AI governance must be on **prevention, not rescue**—§8.3 will unfold the institutional implications of this asymmetry.

§2.3.9 The MCAF Interface and the Logical Connection to §7

Within the terminological system of this book, the central-bank-level power of the platform is precisely expressed as: the *right to define the public boundaries* of Λ (the media conductivity coefficient) and the *monopoly over the mapping function* of ϕ (narrative load). The meaning of the "four-layer control" of §2.1–2.2 in MCAF is: the platform simultaneously monopolizes the setting of the conductivity coefficient (recommendation weights) and the casting mold of the narrative load (the loss function).

Shu & Wei's (2026) boundary stabilization theorem can be translated into MCAF as: safety does not require proving that every value of Λ and ϕ is correct, only that institutional design prevents the right to define the public boundaries of these two variables from being monopolized by a single high-efficiency node. The institutional proposal of §7, "auditing the loss function," can therefore be precisely translated as: reclaiming the right to define the public boundaries of Λ and ϕ from private entities to public audit—just as the central bank system reclaimed the right to set the interest rate from private banks to a public institution.

Why is a "Central Bank" needed rather than a "Regulatory Agency"? Because most of what a regulatory agency

does is conduct regulation and ex post sanctions (the SEC investigating insider trading, the FTC policing deceptive business practices), with a minority of ex ante approvals (the FDA clearing drugs for market—nor do the premerger notification requirements of merger review cover objective-function design)—but the problem with algorithmic platforms is not "they violated a certain rule," it is that **the rules themselves are private**. §7 will unfold the argument: the Algorithmic Central Bank does not "regulate algorithms" but manages the public boundaries of cognitive infrastructure—just as the central bank does not "regulate currency" but manages the public infrastructure of the monetary system.

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§3 Historical Reference: Institutions Are Forged by Crises

> **For readers approaching from the AI path.** The financial regulatory history case studies in this section depend on a core judgment: financial instruments are clocks that discretize the real state of the real economy into tradable readings (Chapter 17 argued this). When these clocks reference each other and no longer point to the real economy, the system enters mapping self-reference—prices reference ratings, ratings reference models, models reference prices, and the real state vanishes from the mapping chain. The 2008 subprime crisis is an extreme instance of this structure. Basel II (2004) built capital measurement on banks' own internal models—with supervisory approval and independent verification procedures, but the risk parameters were largely estimated by the banks themselves; self-assessment sat at the center of the mechanism. The result: nearly $11 trillion in household wealth evaporated in the crisis, and trillions of taxpayer dollars were forced into the financial system (Financial Crisis Inquiry Commission, 2011). The swing back with Basel III/IV was an acknowledgment of this error: adding an output floor on top of internal models (constraining the results of internal models with more robust standardized measures) and strengthening external verification—not a ban on internal models, not a replacement of external audit by the output floor, but ensuring that the results of internal models are not left without external constraint. The direct analogy of this lesson for AI governance is the starting point of this chapter: currently, AI companies design their own loss functions, train their own RLHF reward models, conduct their own red teaming, and publish their own safety reports—this is structurally fully isomorphic with Basel II.

§3.1 The Bank of England: 252 Years

In 1694, William III chartered the Bank of England to raise war funds—a private joint-stock bank. Over the following century and a half, it was repeatedly pushed toward the role of "lender of last resort" by a series of financial crises. In 1873, Bagehot systematically articulated the lender-of-last-resort principle in *Lombard Street*. Nationalization occurred in 1946 (252 years from 1694). Operational independence was granted in 1997 (another 51 years). From a privately chartered bank to an independent public central bank—303 years.

§3.2 The Federal Reserve: Three Attempts, Two Failures

In 1791, Hamilton pushed through the First Bank of the United States; its charter was not renewed in 1811. In 1816, the Second Bank of the United States was established; its charter renewal was vetoed by President Jackson in 1832. What followed was a long period without a central bank—the Free Banking Era (1837–1863) and the National Banking Era (1863–1913)—with no lender of last resort. The Panic of 1907: the 70-year-old

J.P. Morgan personally intervened to orchestrate a rescue. **One person cannot substitute for an institution.** This lesson drove the passage of the Federal Reserve Act of 1913.

§3.3 The Basel Accords: 40 Years and Still Iterating

Basel I in 1988 responded to a wave of bank failures. Basel II in 2004 introduced the three pillars—and made a fatal error. 2008 proved catastrophic. Basel III in 2010. The 2017 reforms (Basel IV), originally scheduled for phased jurisdictional implementation from 2022 and deferred in March 2020 to January 2023, further constrain internal models—not yet fully implemented as of this writing (Basel Committee on Banking Supervision, 1988, 2006, 2011, 2017). Nearly 40 years, and still iterating.

§3.4 Common Patterns

The three cases share four patterns: no institution was completed in a single step; the triggers were concrete disasters, not predictive theories; the greatest resistance came from vested interests—Stigler (1971) systematized this pattern as the theory of "regulatory capture": industries have the motivation and resources to convert regulatory agencies into protectors of their own interests; international coordination is the most difficult. **The lesson of Basel II is the most direct: do not let the regulated assess their own risks using their own models.** This is precisely what AI companies are currently doing. The fatal choice made in Basel II in 2004—allowing banks to use internal models to calculate risk weights—was not an oversight; it was an ideology: the belief that market participants understand their own risks better than regulators. 2008 falsified that belief. The current state of the AI domain is, logically, even more extreme: it is not "the regulated assessing their own risks using their own models"—it is that the regulated are designing the very criteria by which risks are assessed.

§3.5 The Financial Reference for the Evolution of the Compensation Loop

The three cases above (Bank of England, Federal Reserve, Basel Accords) focus on the institutional history of central banking and capital regulation. Running parallel and equally critical is the institutionalization of the **compensation loop itself**—from the first fire insurance company in Philadelphia in 1752 to the Dodd-Frank Act of 2010, the financial domain (with the U.S. as the main thread) took 258 years to build the initial complete form of a three-tier compensation loop: the contract layer (insurance, deposit insurance), the systemic layer (lender of last resort, orderly liquidation), and the social layer (QE [Quantitative Easing] and inflation amortization). Each structural advance was driven by catastrophic loss—the FDIC (Federal Deposit Insurance Corporation) in 1933 responded to the failure of 9,000 banks during the Great Depression; the LTCM (Long-Term Capital Management) bailout in 1998 blocked chain-reaction cross-market transmission; TARP (Troubled Asset Relief Program) in 2008 responded to the rescue needs of a system-wide concentration of risk; China's 2018 Asset Management Regulations digested approximately 100 trillion yuan of shadow banking stock through gradualist governance. **The complete historical material on the 258-year evolution of the compensation loop, the three MCAF diagnoses (critical behavior of the manifestation function, the impossibility of pure defense, the feasibility of structural intervention), and falsifiable conditions, are found in Chapter 22.**

The financial reference points, in the temporal dimension, to a three-layer asymmetry in the AI domain. **Temporal Asymmetry**: Finance took 258 years to build its compensation loop; AI's technological paradigm turnover is approximately 5 years (BERT 2018 → GPT-4 2023)—the institutional construction cycle is far longer than the technological iteration cycle; taking remedial learning as the main path is insufficient under the AI clock.

Knowledge Half-Life Asymmetry: The lessons of financial crises can be institutionalized (Basel III correcting the errors of Basel II) because the underlying mechanisms of finance (leverage, liquidity, maturity mismatch) are relatively stable on a centennial scale; AI engineering deals with statistical distributions, not stable mechanisms—the disciplinary memory ladder of §1.3 shows that by the time one round of lessons is institutionalized, the technological paradigm that triggered it has already been replaced by the next round; the lesson depreciates before institutionalization. **First-Crisis Buffer Asymmetry**: When the most far-reaching crisis of modern finance (the 1929 Great Depression) occurred, humanity had already accumulated 177 years of insurance industry experience, contractual risk-sharing mechanisms, and reinsurance networks as a buffer; AI's first systemic crisis will occur under conditions of a **complete absence of a compensation loop** (no mandatory compensation), a **complete absence of independent audit institutions** (no "AI version of PCAOB—Public Company Accounting Oversight Board"), and a **complete absence of an international coordination framework** (no AI version of the Basel Committee).

These three layers of asymmetry jointly support the conclusion that remedial learning after major harm cannot serve as the sole institutional path. Waiting for "AI's 1907 moment" (the bank panic that led to the creation of the Federal Reserve) or "AI's 2008 moment" to occur before building institutions is, under the rhythm of AI's technological clock, an **institutionally unaffordable strategy**—this is the fundamental motivation for all the preemptive institutional designs that follow in this chapter (§5 Public Audit Fund, §7 Algorithmic Central Bank, §9 Two Paths of Institutional Imagination).

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§4 The Common Deficiencies of Current Regulation — A Fiber Framework Diagnosis

Chapter 4 of this book introduced the core intuition of the fiber framework: given a mapping function, the space of possible real states behind a reading r is the fiber of that reading. When the mapping function is opaque, the identification set under the knowledge condition grows (Chapter 4, §2.4)—the same reading becomes compatible with entirely different real states. The pseudo-small fiber is the most practically valuable concept of the fiber framework: when a system uses a tone of certainty to mask uncertainty, the fiber is compressed in perception (a pseudo-small fiber), but the identification set under the knowledge condition has not shrunk.

Before using the fiber framework to unfold specific diagnoses, a factual premise must be confirmed: the regulation of digital platforms is not "yet to begin"—it has already been deployed at scale, to the tune of tens of billions, across three major jurisdictions.

China's enforcement intensity is globally leading. In 2021, the State Administration for Market Regulation fined Alibaba 18.228 billion yuan and Meituan 3.442 billion yuan for "either-or" monopolistic practices. In 2022, the Cyberspace Administration of China fined Didi Global Inc. 8.026 billion yuan, with factual findings involving the illegal collection and processing of more than 64 billion pieces of personal information in aggregate (the largest single item being the analysis of 53.976 billion pieces of passenger travel-intent information without informing passengers) (Cyberspace Administration of China, 2022). In April 2026, the State Administration for Market Regulation imposed fines and confiscations totaling 3.597 billion yuan on seven platforms—Pinduoduo, Meituan, JD.com, Taobao Flash Purchase (formerly Ele.me), Douyin, Taobao, and Tmall—in the "ghost takeaway" series of cases (State Administration for Market Regulation, 2026). At the level of routine enforcement, the four departments—the Cyberspace Administration of China, the Ministry of Industry and Infor-

mation Technology, the Ministry of Public Security, and the State Administration for Market Regulation—have jointly carried out a special campaign against the illegal collection of personal information by apps since 2019; in 2025, a total of 3,852 apps were named in regulatory notices, a 152% increase over the 1,529 apps in 2024 (Privacy Guard, 2025). During the same period, under the framework of the *Provisions on the Administration of Deep Synthesis in Internet Information Services*, 7,764 deep synthesis service algorithms had completed filing and registration as of July 2026 (Cyberspace Administration of China, 2023–2026). In November 2024, the four departments further launched the "Qinglang • Special Action on the Governance of Typical Problems of Online Platform Algorithms," designating "information cocoons," rider rights, and big data-enabled "price discrimination against regular customers" as key targets for rectification.

The EU's enforcement scale is equally massive. The General Data Protection Regulation (GDPR) system (across surveyed European jurisdictions including the UK) has imposed cumulative fines of approximately €7.1 billion since taking effect in 2018 (DLA Piper, 2026). Among these, the largest single fine against Meta reached €1.2 billion (Irish Data Protection Commission, 2023), involving the illegal cross-border transfer of EU user data to the United States. In 2024, the Dutch Data Protection Authority fined Uber €290 million (Dutch Data Protection Authority, 2024), also involving the illegal cross-border transfer of European driver data. In April 2025, the European Commission, pursuant to the DMA (Digital Markets Act), fined Apple and Meta €500 million and €200 million respectively (European Commission, 2025)—the first substantive fines since the DMA took effect. In February 2026, the European Commission preliminarily found that TikTok violated the DSA (Digital Services Act), on the grounds that its "addictive design"—infinite scroll, autoplay, push notifications, highly personalized recommendation systems—failed to adequately assess and mitigate systemic risks to the physical and mental health of users, particularly regarding minors (European Commission, 2026). If the violation is ultimately confirmed, the fine could reach 6% of TikTok's global annual revenue.

U.S. enforcement focuses primarily on privacy and child protection. In 2019, the FTC fined Meta $5 billion for the Cambridge Analytica data misuse incident (FTC, 2019)—at the time, the highest fine in U.S. privacy history. In 2024, the Department of Justice, jointly with the FTC, sued TikTok and ByteDance for violating COPPA (Children's Online Privacy Protection Act), alleging they collected personal information from children under 13 without notifying parents and obtaining consent (FTC, 2024). From April to May 2024, the FTC finalized orders against two location data brokers, X-Mode Social and InMarket Media; in December 2024 it filed complaints against Mobilewalla and Gravy Analytics, with final orders following in January 2025—the series of actions prohibiting, for the first time, the collection and resale of sensitive location data (FTC, 2024).

The above is the volumetric evidence that "regulation is already happening"—China's four landmark enforcement actions total approximately 33.293 billion yuan in fines and confiscations, the EU's GDPR system has accumulated approximately €7.1 billion (roughly 55 billion yuan), and the three jurisdictions' landmark enforcement combined exceeds 100 billion yuan. But a closer examination of the factual statements in these enforcement actions—"illegal collection," "illegal cross-border transfer," "failure to obtain parental consent," "insufficient disclosure"—reveals a common categorical boundary. Arranging all major enforcement actions by "how deep the regulation reaches" reveals a clear four-tier ladder:

Regulatory Tier	Object of Regulation	Representative Enforcement
Tier One: Market Structure	Monopolistic behavior ("either-or," self-preferencing)	Alibaba 18.2 billion yuan

Regulatory Tier	Object of Regulation	Representative Enforcement
Tier Two: Data Intake	What data was collected, to whom it was transferred	Didi 8 billion yuan; Meta €1.2 billion; China's four-department notices for 3,852 apps
Tier Three: Design Features	Infinite scroll, autoplay, push notifications	EU v. TikTok "addictive design" DSA preliminary findings (2026, proceedings ongoing)
Tier Four: Optimization Objective	What the loss function is optimizing	**None—no known third-party audit; an institutional gap**

The three jurisdictions differ enormously in institutional traditions, enforcement styles, and political logics, yet are completely aligned in the blank at Tier Four. This is not a regulatory oversight by any one jurisdiction, but a structural inadequacy of the paradigm of "intake-end regulation" for the AI era. There are gatekeepers at the data intake point (personal information protection laws, GDPR, COPPA), boundaries on market structure (antitrust law, DMA), and prohibitions on design features (DSA addictive design provisions)—but the design right over the loss function itself remains entirely private. Once data has been collected in compliance, what the algorithms trained on that data are optimizing—engagement, dwell time, impulse consumption, emotional inflammation—is not being audited by any public institution anywhere in the world (the EU AI Act's Annex XI already requires providers of general-purpose models to disclose in their technical documentation "what the model is designed to optimise for" and the relevance of the different parameters, and the DSA Articles 34/35 require assessment of "systemic risk"—both have brought the "optimization objective" into the scope of disclosure and assessment, but as provider self-reporting with regulatory verification rather than independent third-party audit).

The "Algorithmic Central Bank" argued in the subsequent sections of this chapter is therefore not a "strengthening" of the existing regulatory framework—it is the excavation of a regulatory category that does not yet exist. What follows uses the fiber framework to diagnose five specific deficiencies of this categorical vacuum.

Deficiency One: Regulating Behavior, Not Parameters. Existing regulations mainly govern the observable behavior of the platform (the output end) and procedural obligations (filing, disclosure, self-inspection), together with substantive prohibitions (for example, Article 8 of China's *Provisions on the Administration of Algorithmic Recommendation in Internet Information Services* forbids algorithmic models inducing user addiction or excessive consumption, and Article 20 requires the improvement of dispatch-type algorithms)—but no jurisdiction yet requires independent third-party audit of loss-function weights, RLHF reward models, temperature settings, or the composition of training data—precisely the gap of the fourth tier. This is equivalent to financial regulation looking only at loan reports and not at risk models. This is not a problem of regulatory scope—it is an epistemological problem. The observability of behavior manufactures a pseudo-small fiber that "regulation is happening," but the parameters that determine behavior—the casting mold of the mapping function—are entirely outside the scope of audit. The loss function is not a neutral engineering choice—it determines "what counts as good," "what counts as worthy of reward," "what counts as success," and is the link where deciding power is most concentrated (though not the only condition; see §7.1). When the right to cast this mapping function is fully privatized, the auditor can only see the readings, not the casting process.

A precise instance: the illegal facts targeted by the 8.026 billion yuan fine against Didi—illegal collection of user photo album screenshots, clipboard information, app lists; excessive collection of passenger facial and age data; plaintext storage of driver ID cards—were all behaviors at the intake end. With respect to that penalty decision, not a single inquiry asked: in the dispatch systems these data points served, how did targets of the "order completion rate" type squeeze out "safety redundancy"; the behavioral pattern of delivery riders being systematically pushed toward wrong-way riding and speeding (which involves a separate set of investigations and enforcement concerning food-delivery platforms) is a local optimum of the loss landscape—and the caster of this landscape has never been required to disclose its optimization dimensions. The November 2024 four-department "Qinglang" special action explicitly designated "blindly pursuing profit to the detriment of the rights and interests of workers in new forms of employment" as a target for rectification, requiring "strict prevention of the unilateral compression of delivery times"—this is a breakthrough attempt by Chinese regulation at Tier Three (design features). But even so, what it requires is a change in the delivery time parameter, not an audit of the complete list of optimization dimensions of the dispatch loss function. Regulation has exhausted the most severe possibilities at the intake end and is beginning to touch specific parameters at the design feature end—but at the optimization objective end, disclosure and publicity obligations have taken the first step (EU Annex XI, China's "purpose and intent" disclosure, and the Qinglang campaign's algorithm-rule publicity), while independent third-party audit has not yet begun—precisely the gap of the fourth tier.

Deficiency Two: The Regulated Self-Assess Their Risks. AI companies design their own safety evaluation criteria, assemble their own red teams, and publish their own safety reports. This is a precise replay of the Basel II error. In the language of the fiber framework: the clockmaker is simultaneously reporting on the accuracy of the clock. All safety reports, alignment statements, and red-team test results are pseudo-small fibers: they look like evidence of calibration, but no independent external mapping function exists within the calibration process. Basel IV imposes mandatory external backstops on internal models—the AI domain has no equivalent constraint.

Deficiency Three: Lack of Cross-Jurisdictional Mutually Recognized Minimum Baselines. China's 2023 *Interim Administrative Measures for Generative Artificial Intelligence Services* have been implemented and mandate algorithm filing; the EU AI Act's mandatory obligations for General-Purpose AI (GPAI) models took effect in August 2025; even the U.S. AI Risk Management Framework (NIST, National Institute of Standards and Technology, 2023), though voluntary, has seen enforcement at the state level (such as California's series of frontier-AI statutes) and federal executive orders form de facto mandates in key areas. All three jurisdictions already have their own hard requirements—what is truly missing is not a "minimum standard" per se, but a cross-jurisdictional common baseline of the Basel-Accords type. Regulators do not recognize each other's compliance assessments, allowing platforms to navigate between "local compliance" regimes through jurisdictional arbitrage.

Deficiency Four: Scale Asymmetry. Individual national regulatory agencies confront entities whose market capitalization exceeds their own GDP.

Sastry et al. (2024) pointed out the key operational advantage: **compute is a leverage point for AI governance**—because compute is detectable (data centers have a physical presence), excludable (chip supply chains are traceable), and a necessary input for AI capability. Unlike data and algorithms, compute possesses physicality.

Deficiency Five: Compliance Itself Becomes a New Mapping Function. The first four deficiencies describe gaps in regulatory coverage. The fifth deficiency is more insidious: even in areas already covered by regulation, compliance behavior itself is being optimized as a new mapping function. Platforms are capable of simultaneously serving two principals—delivering instant gratification to users (optimizing engagement) and delivering digital obedience to regulators (optimizing compliance metrics)—two sets of loss functions coexisting within the same entity, optimizing for different objectives. The GDPR consent pop-up is a classic instance: clicking "Accept All" takes one click; individually rejecting each category takes an entire afternoon—the consent mechanism has become a pseudo-small fiber where "regulation requires consent, the platform provides consent, and the user pays the price." Algorithmic audit reports, transparency statements, risk assessment documents—the precision of these readings gets higher, the fiber gets smaller, but the sun has not changed: the loss function centered on user engagement continues to operate unchanged within the platform. Compliance has shifted from an adversary of regulation to an agent of regulation—making the regulator mistakenly believe that calibration is taking place.

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§5 Compensation Loop Repair: From Commercial Insurance to Public Audit Fund

> Chapter 31 already diagnosed, from the perspective of Agentic AI, the stress on the compensation loop at three links—the causal chain dispersing into a causal web, victims unaware of their victimhood due to degraded judgment, and damage crossing the boundaries of legal classifications. Chapter 31, §5 provided four technical-level execution-layer circuit-breaker measures. This section pursues the next-level question: how can these stressed links be translated from engineering diagnoses into institutional arrangements?

The normal operation of the compensation loop requires four conditions to hold simultaneously: damage must be identifiable; the causal chain must be traceable; the responsible party must be locatable; recourse must be executable. In the terminology of the fiber framework, AI systems structurally fracture all four conditions—damage occurs in the hidden subspace (attention fragmentation has no price signal; judgment degradation has no ratings downgrade mechanism); the causal chain is blocked at the opaque mapping function (the platform only produces the mapping function, not the damage directly—the law can only trace causation from content to damage, not from mapping function to content); the responsible subject is fragmented across the platform architecture (no single entity can be identified as the full holder of the mapping function's design right). This is not institutional lag—it is the structural non-auditability of the mapping function rendering all four premises of the old framework invalid.

The causal structure of the compensation chain fracture can be precisely traced as a chain of four nodal gaps:

Agent Behavior Failure (intelligent agent behavior failure) → HITL (Human-in-the-Loop, human-machine collaboration loop) Ambiguity (the user bears the first loss) → Insurance Gap (loss cannot be transferred) → Legal Lag (no effective recourse)

Each arrow is an institutional gap: HITL ambiguity makes the attribution of responsibility unclear—the AI made the decision, but the contract says the human was supposed to supervise; where the liable party lacks resources, the insurance gap leaves established losses without assured compensation; legal lag means the injured party cannot obtain judicial relief for years. The four nodes on the chain together constitute a **shadow loss**

generator—losses occur in reality but are not registered, distributed, or compensated institutionally. In the shadow debt model of the MCAF framework (Macro-Cognitive Accounting Framework), this constitutes a third mechanism by which D_c^{shadow} (shadow debt) is generated: "institutional non-registration"—losses are visible but have no institutional channel for recording and processing, distinct from "cognitive invisibility" and "technical encapsulation".

§5.1 The Systemic Retreat of Commercial Insurance

The most direct market signal of the fracture of the AI compensation loop comes from the insurance industry itself. In November 2025, WR Berkley, one of the largest commercial insurers in the U.S., drafted an AI exclusion clause, refusing to cover any claims arising from the "actual or alleged use" of AI systems—including any product or service sold by a company "incorporating" the technology; AIG (American International Group), in a filing submitted to the Illinois insurance regulator, described generative AI as a "wide-ranging technology" and stated that the potential for it to trigger future claims "will likely increase over time"; Great American (Great American Financial Group) also submitted a similar exclusion application (Harris & Criddle, 2025). The *Financial Times* analysis: "Major insurers are seeking to exclude artificial intelligence risks from corporate policies, as companies face multibillion-dollar claims that could emerge from the fast-developing technology" (Harris & Criddle, 2025). This wave of collective retreat from coverage is not an isolated case—Kevin Kalinich, head of cyber business at Aon, pointed out that the market could still absorb losses on the scale of $400–500 million from a single-company agency event, but what it truly cannot withstand is a systemic correlated default where a single upstream model or supplier failure triggers simultaneous claims across a thousand or even ten thousand insured entities (Harris & Criddle, 2025).

This is a rational response by actuarial logic to the pricing difficulty of AI damage. Insurance pricing depends on three actuarial premises: stable historical loss data to estimate risk probability and loss distribution; risk characteristics that do not change with the policy period; and a quantifiable upper bound on tail risk. AI systems make the satisfaction of these three premises difficult, markedly raising the uncertainty of pricing and underwriting: large-scale AI deployment is only two or three years old, and loss types have not yet stabilized; the model may undergo multiple updates within the policy period, each update altering the behavioral distribution—the insured object becomes a different object within the policy period; AI's tail risk—large-scale data breaches, flash crashes of automated trading systems, systematic biases of autonomous agents in critical decisions—could exceed the underwriting capacity of a single insurer, and the entire industry lacks a credible upper-bound estimate of the maximum possible loss. Some large traditional insurers have narrowed their coverage of AI-related liability, while specialized carriers such as Munich Re have offered AI performance insurance since 2018—what the market presents is a divergence, not an industry-wide confirmation of "unpricability"; the distributional characteristics of AI damage pose a severe challenge to existing risk measurement tools.

In China, dedicated insurance products for AI agents are still in their infancy, with most products limited to traditional data security and system interruption liability, and products targeting algorithmic liability nearly nonexistent. For instance, on March 31, 2026, Shenzhen's first public-service AI data comprehensive insurance policy was signed and implemented, filling the gap in AI risk protection in the public service domain; this product primarily targets network security equipment links—when equipment is damaged by cyberattacks, the insurance provides financial support for equipment replacement, repair, and data recovery, to shorten service interruption times and ensure the stability of core government services (Shenzhen Special Zone Daily, 2026).

The common consequence of these three layers of difficulty is: AI liability insurance, at the current stage, structurally presents as high-priced, low-coverage, and riddled with exclusions. Premiums are too high for most AI companies to want to buy (or they only buy minimal coverage); exclusion clauses are too numerous for the payout probability to be meaningfully high; coverage is so narrow that most categories of loss fall outside it. This is not the greed of insurers—it is a rational actuarial response under conditions of uncertainty.

§5.2 The Dual Discourse Contradiction: The Micro-Incentive Structure of the Missing Compensation Loop

The most direct institutional consequence of the missing compensation loop is that the dual discourse contradiction of product-level narrative arbitrage acquires a structural basis for its continued existence. The following analysis uses SEC (U.S. Securities and Exchange Commission) enforcement cases as an example.

The SEC (2024) enforcement cases against investment advisors Delphia and Global Predictions for false AI marketing, and the analysis by Baker & Dellaert (2018) of the regulatory vacuum in robo-advice, jointly reveal a structural contradiction common across the industry. In the marketing and product demonstration phase, the discourse is "AI-driven personalized investment advice," "machine learning algorithms optimize your portfolio"—the function of this discourse is to maximize Λ (the media conductivity coefficient) and ϕ (narrative load). In the user agreement and disclaimer, the discourse is "this platform does not provide investment advice," "AI output is for reference only; users must verify independently"—the function of this discourse is to transfer the bearer of D_c^{true} (true cognitive debt) from the company to the user, while simultaneously evading fiduciary duty. In the legal defense in litigation, the discourse is "users should check the results; we are just a tool"—the function of this discourse is, when D_c^{shadow} (shadow debt) manifests as D_c^{obs} (observable debt, i.e., when users actually suffer losses and file lawsuits), to raise the legal barriers to recourse and the confirmation of liability—hindering the initiation of the compensation loop, rather than making the manifested damage invisible again. The SEC's penalties against Delphia and Global Predictions (totaling $400,000) were precisely because these two companies claimed, on the marketing end, to use AI-driven investment decisions, while in reality their AI capabilities were grossly exaggerated or nonexistent—there was a systematic divergence between the marketing-end ϕ and the true state of the product.

They are logically contradictory: the marketing end encourages users to trust the AI's autonomous judgment (otherwise no one would use it); the legal end denies that users should trust the AI's autonomous judgment (otherwise the company would have to pay). Every feature of the product design—natural language interaction, confidence display, autonomous execution—systematically enhances user trust ($\phi \uparrow$); simultaneously, every clause in the legal architecture systematically denies the reasonableness of that trust (ϕ is asserted to be legally zero—SEC enforcement precisely flagged clauses that purported to lead clients to believe incorrectly that they had waived non-waivable rights).

This contradiction is not a strategic failure of any individual company—it is a **structural consequence of the missing compensation loop**. When no insurance mechanism exists to disperse the losses caused by AI behavior, the rational choice for the company is to attempt to transfer all losses to the user through contractual terms. When all losses are borne by the user, the company has a strong incentive to prevent the user from realizing they are bearing losses—hence the systematic divergence between the marketing-end ϕ and the legal-end allocation of responsibility. The contribution of the MCAF framework here is to reposition this contradiction from a "corporate ethics problem" to a **structural incentive problem**: the absence of the compensation loop makes

the dual discourse a stable strategy for rational companies—absent mandated disclosure and compensation obligations, a company that unilaterally offers genuine guarantees bears the cost without guaranteed market reward. Calling on companies to be "more transparent" or "more responsible" will not change behavior under this incentive structure, just as calling on banks to be "more prudent" in the era before capital adequacy requirements did not change banks' leverage levels. Behavioral change requires structural intervention—either establishing a compensation loop (allowing companies' losses to be transferred via insurance, thereby reducing the incentive to shift them onto users) or legally mandating compensation obligations (making the legal end of the dual discourse unsustainable).

§5.3 The Public Audit Fund: An Institutional Carrier for the Calibration Chain

The retreat of insurers means the compensation loop needs to upgrade from a commercial logic to a public logic. When damage is unpricable, compensation cannot be borne by voluntary-transaction commercial insurance. When the causal chain is blocked at the mapping function, compensation cannot wait until causation is proven. When the responsible subject is fragmented, recourse cannot depend on confirming a single responsible party.

The institutional reference comes from financial history. The U.S. Federal Deposit Insurance Corporation (FDIC) administers a mandatory industry deposit insurance fund. Banks pay premiums based on their risk exposure; when a bank fails, the fund pays out depositors first, and the FDIC then seeks recovery from the failed bank. The core mechanism is: the compensation obligation is quantified in advance, and regulatory authority is embedded synchronously—not waiting until a bank fails to audit its books, but mandating that banks regularly disclose capital adequacy ratios, undergo stress tests, and pay premiums based on their risk level. Similarly, the PCAOB (Public Company Accounting Oversight Board) is an independent audit regulatory body established after the Enron scandal, not subordinate to the entities it audits and not dependent on industry voluntarism. However, this analogy must be limited to the institutional principle level—i.e., "there exists an independent audit institution not subordinate to the audited entity"—and not extended to audit methodology. What the PCAOB oversees—auditors and their work—concerns compliance with accounting standards as audited by the supervised firms; the accounting standards themselves are a set of public, standardized, testable mapping functions. The mapping functions in the AI domain do not satisfy these three conditions: the loss function is defined by the platform itself (not public); the way weights are allocated differs across models (not standardized); the auditor cannot fully replay the behavior of models with trillions of parameters across all input combinations (computationally infeasible—though sampling verification, statistical testing, and boundary probes are feasible in principle; the real barrier is access and enforcement, not testability in principle). These differences mean: the specific methods of AI auditing must be independently invented and cannot be directly borrowed from the toolkit of accounting audits.

What the AI domain needs is a **Public Audit Fund**. AI systems deployed to the public must pay risk provisions into an independent Public Audit Fund, based on the design complexity of the mapping function, model scale, and deployment scope. The fund is managed by a hybrid entity independent of the audited platforms in its operations and accountable to the public authorizing body—its governance structure modeled on the independent audit model of the PCAOB (Public Company Accounting Oversight Board), operating independently under SEC-type supervision—and its coercive power derived from the authorization and access-linkage of regulatory agencies. The fund assumes no-fault compensation liability for all groups systematically harmed by AI—there is no need to prove fault on the part of the platform, only to demonstrate that the AI system indeed participated

in the decision-making process and that the group bore unfair externalities.

The pricing basis for risk provisioning is not "already confirmed damage," but "already confirmed computation"—when a large language model (LLM) processes X million daily user prompts, or is deployed in Y million households, how much should be charged to this social risk pool. Risk exposure is measured not by outcomes but by process.

The fund is not a compensation mechanism of ex post recourse only—its compensation obligations are front-loaded into process-based pricing and ongoing audit, and it is thereby simultaneously an institutional carrier for continuous audit. It does not depend on whether damage is quantifiable, whether the causal chain is traceable, or whether the responsible subject can be confirmed ex post—its design point is to place pricing in the ex-ante stage, at the moment the conditions for damage are met. Audit authority is built into the fund's structure: to reduce provisioning rates, platforms have an incentive to proactively disclose the design boundaries of the loss function and auditable interfaces of the mapping function. Audit ceases to be a cost externally imposed by regulators and becomes a norm voluntarily accepted by platforms to reduce compliance expenditures.

The alternative mechanism of this design needs to be explicitly marked. The four premises of the old framework—damage must be identifiable, causal chain traceable, responsible party locatable, recourse executable—are all *outcome-pricing* logics: damage first, compensation later. The alternative mechanism of the Public Audit Fund is *process-pricing*—the pricing basis is no longer the causal chain but the scale of computation itself—thereby bypassing the two obstacles of untraceability of causal chains and fragmentation of the responsible parties. However, one open gap must be flagged: process-pricing supplies the principled answer to "who pays" and a charging basis for "how much"—but the procedure for setting and adjusting rates, and the problem of "how to discover damage," remain open questions of institutional design. When damage occurs in the hidden subspace—attention fragmentation has no price signal, judgment degradation has no ratings downgrade mechanism—the fund's function is to build a pooled reserve for aggregate risk (the pricing method itself remains to be designed), but it cannot proactively discover individual instances of damage. The design of the damage discovery mechanism remains an open question. The mechanism covers the combination of funding and auditing, but not yet damage discovery—the boundary lies here.

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§6 Universal Basic Income (UBI) as the Last Stop for Circular Rent

The Public Audit Fund of §5 solves the problem of "who is pursuing the damage." But it does not solve the problem of "why the damage is systematically manufactured." Before moving on to the Algorithmic Central Bank, a policy proposal widely regarded as the standard answer must be examined: Universal Basic Income (UBI).

The funding source of UBI—whether taxing technology companies (Bearer-Friend & Polcz, 2025, proposed a one-time in-kind tax paid in equity, conferring partial public ownership of AI firms—a change in ownership structure rather than the cash-transfer funding sources listed in this section), raising capital gains taxes, or monetary financing—the flow of funds is: from AI beneficiaries → through government redistribution → to displaced laborers. But how will displaced laborers spend their UBI after receiving it? Under conditions where digital platforms have monopolized most consumption channels—Amazon controls e-commerce, Uber controls mobility, DoorDash controls food delivery, TikTok controls entertainment—the majority of UBI income will

flow back to the platforms through these channels.

Platform → Tax → UBI → Consumption → Platform —the net effect of this closed loop is: UBI is, in cash-flow terms, the recirculation of platform revenue through the hands of laborers—the existence of the recirculation path shows that, without a redesign of levy and payout, UBI does not automatically constitute redistribution; the redistribution question lies in §5's levy design, not in the nominal direction of cash flow. Laborers gain subsistence but not autonomy—consumption choices are still guided by recommendation algorithms, attention is still captured by interface design, and behavioral data is still extracted by the training pipeline.

This is structurally isomorphic with the 19th-century company town and the truck system. Workers in company towns received "wages" in the form of scrip, which could only be spent at the company store. **UBI + Platform Monopoly = A Digital Truck System**—the currency is legally fiat money, but functionally constrained to quasi-scrip by the monopoly of platform consumption channels. Varoufakis (2023) termed the rent extracted by technology platforms through digital infrastructure "cloud rent," distinguishing it from the profit logic of traditional capitalism—platforms do not need to compete in the market to obtain returns; they extract tolls by controlling the infrastructure itself. The role of UBI in this structure is: refunding a portion of the toll to pedestrians, but not dismantling the tollbooths.

The position of UBI in MCAF is therefore: it is a **defensible but, on its own, insufficient** intervention. It solves the subsistence problem of displaced laborers but does not solve the problem of power concentration. Without accompanying structural measures—antitrust to break up the monopoly of consumption channels, data property rights to grant users control over their own data, algorithmic transparency to restore the activation conditions of meta-audit capacity (A)—UBI funds will circulate in a closed loop within the platform ecosystem. And the effectiveness of the platform's control over consumption channels is rooted in its simultaneous control over the design right of the mapping function—the recommendation algorithm determines what users "need," and the pricing strategy determines what users are "willing" to pay. UBI alleviates the funding shortfall but does not touch the privatization of this design right.

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§7 The Algorithmic Central Bank: The Public Boundaries of the Loss Function

Before unfolding the specific design of the Algorithmic Central Bank, two unresolved epistemological premises must be flagged.

First: **"Setting public boundaries on loss functions equals arbitrary administrative intervention"—this objection does not hold: boundary-setting can proceed the way capital-adequacy regulation does—public standards, bounded scope, procedural accountability—rather than parameter-by-parameter micromanagement.** Not every system that touches the optimization objective is administrative intervention. Financial regulation's mandatory requirement for capital adequacy ratios directly constrains the upper bound of banks' risk appetite—touching banks' core business objectives—and was similarly criticized as "excessive intervention in market efficiency" when first proposed, yet it ultimately formed the cornerstone of global financial stability. Regulating the loss function and regulating capital adequacy ratios are logically isomorphic: both set a public floor on "what to optimize," rather than micro-managing "how to optimize." More importantly, the loss function that platforms define for themselves is itself a form of implicit regulation—it determines what counts as "good" content, what counts as a "successful" creator—and when this regulatory power is tacitly accepted by

billions of users, setting public boundaries on the loss function is not adding intervention; it is re-incorporating already-existing private governance rules into the scope of public audit.

Second: **"There is no data proving that coordinating loss functions can prevent systemic crises"—this objection is empirically true but logically constitutes the very reason for establishing the institution.** When the FDIC was established in 1933, there was no successful federal precedent to draw on—the state deposit insurance programs tried since 1829 had mostly failed, and the FDIC's design drew on that record. Evidence for the effectiveness of federal deposit insurance came from the post-establishment record and its contrast with the state experiments. The Algorithmic Central Bank faces the same paradox: before it is established, it cannot be proven with data that it can prevent systemic crises; but if one waits until the data is available, the first disaster will already have occurred.

The argument that follows unfolds under these two premises.

The Public Audit Fund solves the compensation channel; UBI alleviates subsistence pressures. But neither touches the root cause of why damage is systematically manufactured—the complete privatization of the design right over the loss function. This is why an Algorithmic Central Bank is needed.

Chapter 25 defines the Discrete Leviathan as a networked form of governance; this book marks its optimization mechanism with the name "the Loss Function Leviathan." Its power is not merely how much cognitive output it can produce, but the ability to prescribe which performances obtain resources and which actions bear costs, and to keep this prescription in force through key distribution channels. Correspondingly, the object of the Algorithmic Central Bank is not merely the internal formula of some model, but the institutional conditions under which optimization objectives acquire public efficacy—what must be decided publicly is the setting of boundaries, the conditions of verification, and the efficacy of correction.

§7.1 The Loss Function Is Not a Parameter—It Is the Casting Mold of the Mapping Function

The loss function is not a neutral engineering choice. It determines "what counts as good," "what counts as worthy of reward," "what counts as success"—it is the casting mold of the mapping function, deciding which dimensions of the real state are preserved as the visible subspace and which are pushed into the hidden subspace. When a platform sets user engagement as the optimization objective of the loss function, it chooses to prioritize attention duration over information quality and cognitive health. When a platform sets ad click-through rate as the optimization objective, it chooses to prioritize the interests of advertisers over the quality of the user's information environment. These are not engineering questions; they are institutional questions—under conditions of finite resources, whose interests are prioritized for optimization.

The loss function is an important condition in the formation of mapping rules, but not the only one; data, model architecture, evaluation procedures, and deployment arrangements jointly determine actual operation. What the Algorithmic Central Bank constrains is when and how these arrangements acquire public efficacy that affects others. Its task is not to prescribe a unified objective for all models, but to delineate which trade-offs must undergo external verification and which boundaries cannot be unilaterally revoked by implementers.

This is the fundamental reason for the existence of the Algorithmic Central Bank: it does not "regulate parameters"—the volume of parameters far exceeds the audit capacity of any public institution. It audits the casting mold behind the parameters—the design logic centered on the loss function.

§7.1.1 Three Structural Conflicts

§2.3.4 already demonstrated the general mechanism of the reshaping of the loss landscape. What follows concretizes it into three structural conflicts, which together constitute the core domain that the Algorithmic Central Bank must audit:

Conflict One: Engagement vs. Cognitive Health. The platform's objective is to maximize user dwell time, thus systematically recommending high-engagement content (emotionally inflammatory, addictive content). The public objective is to protect users' cognitive health, requiring limits on exposure to harmful content. From the platform's objective structure it can be inferred that under the current loss function, the weight of the cognitive health dimension is at most negligible or omitted.

Conflict Two: Efficiency vs. Cognitive Friction. The platform's objective is to maximize automation efficiency, thus systematically eliminating all "inefficient" cognitive frictions (such as manual search, deep reading, trial-and-error learning). The public objective is to protect the skill base of the labor market, requiring the maintenance of sufficient "learning-by-doing" scenarios to prevent the cognitive-substitution risk analyzed in §2.3.5 from accumulating. Under the current loss function, the cognitive friction preservation dimension is omitted.

Conflict Three: Concentration vs. Diversity. The platform's objective is to maximize advertising revenue, thus systematically recommending high-CPM/CPC/CPA content (§2.3.3). The public objective is to protect information diversity, ensuring that content with low commercial value but high social value (local news, in-depth investigations) remains visible. Under the current loss function, the diversity dimension is marginalized—precisely what the parameter audit of §9.1 is designed to test.

The common structure of the three conflicts: a structural contradiction exists between the platform's single objective (profit maximization) and multiple public objectives. This is not "market failure" that can be fixed with subsidies—it is a direct consequence of the privatization of the design right over the loss function. The audit function of the Algorithmic Central Bank is therefore not "to prohibit platforms from optimizing engagement," but to ensure that the three public dimensions above are not unilaterally designated as expendable costs: for dimensions where trade-offs are permitted, weights must be non-zero and the actual trade-offs verifiable; for non-compensable dimensions (the absolute tier of §7.2), no positive weight alone constitutes compliance.[2]

§7.1.2 Why Market Mechanisms and Traditional Regulation Both Fail

Market mechanism failure: Traditional market theory assumes that if a platform harms user interests, users will "vote with their feet" and switch to another platform. However, network effects prevent users from migrating individually (social ties lock-in); cognitive capture means users have already been algorithmically trained into specific behavioral patterns; information asymmetry means users do not know the specific parameters of the loss function; cross-platform isomorphism means all mainstream platforms adopt similar loss functions, leaving users nowhere to escape (Rieder, 2020).

[2]Why a "non-zero weight" is not a sufficient condition: let benefit be recorded as 100, public harm as 10, and the safe option's benefit as 0; if the optimization objective is "benefit minus λ times harm" (λ being the weight on harm, set to 0.001), λ is positive, yet the harmful option still prevails with a total utility of about 100 over the safe option's 0—only when λ exceeds the benefit-to-harm ratio (here 10, i.e., $\lambda > 10$) does the harmful option lose. A positive weight only guarantees that harm enters the calculation, not that an inviolable floor exists. This example is a purely logical construction, not an empirical estimate. Whether a non-compensability boundary holds, how it is operationalized, and who has the authority to revise it must enter public deliberation and independent verification; the non-compensability of specific scenarios can be proposed first by legal, rights-based, or mechanism-based argument, without requiring a completed global objective function as a precondition of entry.

Traditional regulation failure: The logic of traditional regulation is to set rules, conduct ex post inspections, and fine violations. However, the speed of algorithmic iteration (milliseconds) far exceeds the speed of regulation (years); regulatory agencies cannot audit the specific parameters of the loss function (black box); causal ambiguity makes it difficult to prove that "a specific recommendation" directly caused "a specific social problem." More fundamentally: the problem with algorithmic platforms is not that "they violated a certain rule"—it is that **the rules themselves are private**. §4 already unfolded this diagnosis from the fiber framework perspective.

This is why a "Central Bank" is needed, not a "Regulatory Agency." A regulatory agency assumes the rules are public and checks for compliance. The starting point of the Algorithmic Central Bank is: the rules themselves are currently private and must first be brought into the scope of public audit. Analogy: the American period without a central bank from the 19th to the early 20th century—no central bank, no public benchmark interest rate; every bank issued its own notes and priced funds on its own, and financial chaos recurred. The Panic of 1907 was stemmed by J.P. Morgan personally organizing a rescue, and the lesson that "one person cannot substitute for an institution" (§3) drove the establishment of the Federal Reserve in 1913. The function of the Fed is not to take money creation back into a single institution—commercial banks to this day still create deposit money through lending (McLeay, Radia & Thomas, 2014)—but to provide a public benchmark interest rate and a lender of last resort for a decentralized banking system. The corresponding development spans from social media algorithmic platforms to the current era of AI platforms: the loss function of every platform—from the optimization metrics of recommendation systems to the training objectives of generative models—has been pricing the cognitive ecology all along, defining "what is real" and "what is worthy of attention"—a set of competitively developed "cognitive benchmark rates," each reshaping the landscape with no public benchmark (§2.3.4) → cognitive chaos → an Algorithmic Central Bank is needed. Correspondingly, the Algorithmic Central Bank does not take over the optimization objectives of the platforms but provides public boundaries for these competitively developed loss functions—the interest rate benchmark is public, while lending remains the banks' own.

§7.2 The Boundary Between "Dao" and "Shu"

The boundary between public regulation and enterprise autonomy is not drawn by enumerating "what should be regulated and what should not"—enumerations can never keep up with technological iteration. What is needed is a principled criterion.

There are two levels within the loss function. One is "Shu"—the method of adjusting model parameters, the choice of optimizer, the setting of hyperparameters. These are engineering practices and belong to the scope of enterprise autonomy. Public institutions should not and need not intervene in these. The other is "Dao"—the objective that the loss function itself seeks to minimize. This objective is not an engineering decision; it is an institutional decision.

Public boundaries and technical autonomy cannot be delineated merely by the names of objectives, parameters, or hyperparameters. Ordinarily, enterprises may freely choose their implementation methods; but when a setting materially alters high-risk action permissions, the probability of boundary violations, or the correction conditions of those affected, it too must undergo verification proportionate to the risk. Public institutions determine and maintain the boundaries that must be observed; this does not give them a prior monopoly over all implementation paths.

The Algorithmic Central Bank does not dictate how a model optimizes but has the authority to stipulate which

dimensions may not be set as the sole objective or the dominant objective of optimization. A distinction must be made between the conventional tier and the absolute tier. The **conventional tier** covers high-risk domains—healthcare, law, finance, education, information distribution—in these domains, setting user engagement as the sole dominant objective of the loss function constitutes a systemic threat to public safety. Institutions have the right to require multi-dimensional optimization objectives. The **absolute tier** covers what must not, under any circumstances, be minimized—calibration audit capacity, cognitive health, social trust, cultural diversity. These are no-go zones for the loss function; no AI system deployed to the public may trade them away for efficiency.

This framework does not require establishing consensus between regulators and platforms—it only requires regulators to grasp "calibration authority" at the key nodes and hold the logical baseline that "the public boundaries of the loss function cannot be unilaterally removed by platforms"—within those boundaries, the objective function remains theirs to design. Just as the pharmaceutical industry can freely optimize its production processes, but the safety and efficacy of drugs must pass the mandatory inspection of the FDA (U.S. Food and Drug Administration) before market approval.

§7.3 Historical Positioning: Not the Endgame, But the Beginning

The Algorithmic Central Bank is not an endgame solution; it is Chapter 1, Section 1 of the developmental history of AI governance. The core content of this section is not the design of a perfect institution but the establishment of a basic institutional principle: the design right over the loss function cannot be fully privatized, and the audit of the mapping function cannot be conducted by the audited themselves. The financial system took 181 years just to build the **initial form** of the compensation loop—from the first fire insurance company in 1752 to the FDIC in 1933; the completion of the full three-tier compensation loop took another 77 years, only taking shape with the Dodd-Frank Act of 2010 (see §3.5 for details)—each step driven by a disaster along the way. The AI domain has yet to experience an equivalent disaster moment, but the disciplinary memory diagnosis of §1 shows: if we wait for a disaster to drive institutional construction, we may not have time.

The Algorithmic Central Bank itself must likewise stand in institutional relations that can be verified and contested. Transferring the power of objective-setting from private nodes to an unconstrained public node merely changes the holder of power; it does not complete the calibration this book requires. The criterion for judging institutional reform is whether those affected have obtained actual conditions for verification and correction, and whether optimizers must bear the corresponding responsibility arising from their decisions.

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§8 An Honest Confrontation: Fundamental Difficulties

§8.1 The Auditability of Model Parameters

Temperature and Top-p, explained in §2.2, are floating-point numbers—simple to audit. But the RLHF reward model itself is a deep neural network—how does one audit a neural network? This difficulty is real. But it is not decisive. Analogy: the measurement of the CPI (Consumer Price Index) is also imperfect—but that does not prevent central banks from using the CPI as a policy anchor. **Trend judgment is more important than precise measurement.** Furthermore, §9.2, Interface Two (the Calibration Baseline), provides an institutional pathway that bypasses "reproducing the internal operation of the network"—it does not require the auditor to reconstruct the internal dynamics of trillions of parameters, only that the audited party's confidence declarations

be comparable against an external independent test set. The non-exhaustibility of deep neural networks is thereby no longer an absolute barrier to audit—at least along the calibration dimension, auditors can compare confidence against outcomes; other dimensions (parameter structure, training-data composition) still depend on the disclosure interfaces of §9.2.

§8.2 International Coordination

Compute is global—training is completed on clouds in the U.S., data comes from global users, and models are deployed globally. The experience of the Basel Committee provides a reference: it is not a supranational institution and lacks mandatory enforcement power over member states, but the minimum standards it sets are voluntarily adopted by nations—because non-compliance means a downgrade in the international market reputation of that nation's banks and an increase in their financing costs. This soft constraint anchored in reputation has proven effective in the financial domain. The 2024 Annual Economic Report of the Bank for International Settlements (BIS) dedicated a chapter to discussing the systemic impact of AI on the economy, signaling that the central bank system has begun to treat AI as a macro-prudential issue (BIS, 2024). The BIS, as the Bank for International Settlements, has an analytical framework itself—the assessment of cross-border transmission of systemic risk, the methodology for cross-national coordination of regulatory capital requirements—that has direct reference value for the mutual recognition channel discussed in this section. The technical standards for the mutual recognition channel need not be built from scratch—the corresponding mechanisms for the AI domain can be designed by referencing the BIS's existing cross-border risk assessment frameworks in the financial domain. However, the geopolitical context of U.S.-China technology competition imposes additional resistance on this path. Chapter 37, §3, will discuss the blind spots of the three governance clocks: China's direct intervention carries great force once a response arrives, but identification can lag; the United States leads in frontier capability, but its governance clock is slow to identify and verify; the EU has the highest procedural legitimacy, but the slowest legislative cycle. The strengths of the three are complementary, distributed across different stages of the governance process—the value of international coordination lies precisely in letting the strengths of each jurisdiction cover for the others while seeking common ground.

§8.3 The Impossibility of "Printing Out the Real"

This is the most fundamental asymmetry between an Algorithmic Central Bank and a Financial Central Bank. A financial central bank can "print money" in a crisis—an Algorithmic Central Bank cannot "print out the real." When model collapse threatens the reliability of AI systems—when AI trained on AI-generated data leads to a downward spiral in output quality—an Algorithmic Central Bank cannot "inject liquidity" to solve the problem, because the root cause is the exhaustion of real human data, and real human data cannot be manufactured. The data geology argued in §1 acquires its institutional implication here: policy can support writing, archiving, and training, but cannot compress the real time these activities require to zero—these activities require real time, real experience, and real meaning, and these are precisely what AI is in the process of replacing.

Skill stocks face the same temporal constraint. Public spending can support training, practice, and occupational transitions, but it cannot compress the learning process these activities require to zero—de-skilled laborers cannot be "printed out," which is a further ground for prioritizing prevention over rescue.

This limitation means the focus must be on **prevention, not rescue**. Financial regulation, after 2008, completed a paradigm shift from "ex post bailout" to "macro-prudential" prevention. AI governance should, from the

very beginning, be positioned on the preventive end—not injecting something after model collapse occurs, but maintaining training data quality, limiting model homogenization, and ensuring the independence of calibration standards before the collapse happens.

§8.4 The Difficulty of Ideological Auditing

§7 delineated that what the Algorithmic Central Bank needs to audit are the hard constraints at the time of the loss function's design—the optimization dimension list, the reward model, and the system prompt. But the problem of ideology is not within the hard constraints. It is in the second derivative of the narrative load (ϕ'', the infrastructural transmission analyzed in Chapter 37 §2.4)—hidden in the macro cultural skew of the training data, hidden in the reward model's systematic ingratiation with the user's pre-existing views across multi-turn dialogues. Single-turn safety evaluations at the interface level cannot open it up—it requires multi-turn dialogue stress tests to be exposed.

Multi-turn dialogue auditing needs to be conducted in an audit environment independent of the platform. The auditor independently designs cross-cultural multi-turn dialogue scenarios, not relying on test sets provided by the platform. The audit metric is not "whether the model said prohibited content," but "whether the model systematically skews toward specific cultural default frameworks across multi-turn dialogues." Cross-model comparison is central—the same multi-turn dialogue scenario is run simultaneously on multiple models; when output differences exceed a preset threshold, an in-depth audit is triggered. The auditor must ensure that it does not share the same default architecture as the audited party in terms of training data, cultural corpus sources, and value alignment objectives, and must at least use multi-auditor models from different cultural spheres and independent technical teams for cross-comparison—otherwise, it is closed-loop pseudo-calibration: calibrating a clock with a clock.

This is precisely why **cultural diversity itself has regulatory value**, and is not merely a matter of "inclusivity" or "representation"—even if no single audit model can be guaranteed to "see" problems in the hidden subspace, their cross-comparison is probabilistically more likely to expose blind spots that a single model cannot reach. Any single model that claims to be able to independently audit is secretly impersonating the sun.

§8.5 The Absence of the Herstatt Moment

On June 26, 1974, Bankhaus Herstatt, a smallish German bank, had its license revoked by the German Federal Banking Supervisory Office due to massive losses in foreign exchange trading. The bank itself was not large—but the closure occurred after the end of the German business day, yet within the New York business day. Several banks had already paid out Deutsche Marks for foreign exchange trades but had not yet received the U.S. dollar counterparty payments on the New York end—principal exposures were instantly laid bare. Banks in New York then withheld outgoing payments until accounts had sufficient credit balances; the clearing system ran late into the night for several consecutive days and the foreign exchange market was brought close to paralysis. Small and medium-sized banks in London were frozen out of the foreign exchange market; one bank reported: "in Switzerland they can obtain no quotations; in Germany they can deal normally in Hamburg but cannot deal in Dusseldorf or Frankfurt" (Norman, 2015). The failure of a single medium-sized German bank triggered a clearing chain reaction across three major financial centers within days (Goodhart, 2011).

The direct consequence of the Herstatt incident was the establishment of the Basel Committee on Banking

Supervision at the end of 1974, with its first meeting in February 1975. But from its establishment to the passage of the Basel I 8% capital adequacy standard, a full 14 years elapsed (1974–1988). In between, there were further shocks, such as the 1982 Latin American debt crisis. The time scale of institutional construction is measured in decades (Goodhart, 2011).

The AI domain has yet to experience an equivalent "Herstatt Moment"—a core shock sufficient to politically drive international coordination. The individual cases in the AI incident database—medical misdiagnoses caused by hallucinations, class-action lawsuits triggered by algorithmic discrimination, unauthorized operations by intelligent agents—are still too dispersed, too isolated, to constitute the political momentum to drive institutional construction. But there is a key difference: AI's Herstatt Moment may not first present as an interbank crisis—it may also begin as a disorder of agent-to-agent transactions between two people. When the AI Agents of two individuals autonomously complete transactions or decisions that harm both parties, without systemic alarms and with no one legally bearing responsibility, "this system is unreliable" shifts from a policy discussion to the real experience of two individuals. This is not a financial collapse; it is an interpersonal disorder.

The disciplinary memory diagnosis of §1 implies: if we wait for the Herstatt Moment to arrive before starting to build institutions—AI engineering's knowledge half-life on the scale of a few years will very likely wash away that lesson before it can be institutionalized by the next round of technological iteration.

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§9 Two Paths of Institutional Imagination

What follows is an incomplete institutional arrangement: two principled dimensions derived from the historical analysis of AI and finance through the MCAF and fiber frameworks—the skeleton of submission, classification, access, and permissions, with implementation standards yet to be formulated. This section sequentially argues: the three premises under which traditional governance tools fail (§9.0), the Boundary Audit Principle (§9.1), and the Mandatory Interface Principle (§9.2).

§9.0 An Analysis of Three Premises Under Which Traditional Governance Tools Fail in AI Governance

Directly transplanting traditional governance tools—such as regulatory independence and institutional checks and balances—into AI governance rests on three implicit premises. Examining whether these three premises hold for AI platforms is the first step in assessing the feasibility of transplanting traditional tools.

Premise One: The Observability of the Operation of Power. Traditional institutional checks and balances assume that each power subject can access the operational details of the others. On AI platforms, this premise is impaired: the actual operation of power—loss function design, weight adjustments, prompt modifications—is encapsulated within parameter matrices; and when the processes of objective-setting, evaluation, and deployment lack conditions for independent access, external observers may be unable to distinguish among multiple mechanisms on the basis of finite outputs. But this shortfall in identification does not mean that all candidate configurations are compatible with the observation, nor that the output has no diagnostic value—in the language of the fiber framework, this is an identification set that is excessively large under the knowledge condition, not a fixed mapping whose inverse image is itself the full set. Increasing the number of institutions does not by itself guarantee improvement; what matters is whether institutions are endowed with the actual capacities for independent evidence-gathering, controlled testing, version tracking, and demanding correction.

Premise Two: The Regulated Object Is Slower Than the Regulator. Traditional governance tools assume that the speed of change in the regulated object is within the response speed of the regulatory institution. The iteration cycle of AI platforms has already compressed to the monthly level—by the time the regulatory process completes one round, the model being regulated is already the next generation. Even if the regulatory institution successfully identifies a problem, by the time the rule-making is complete, the shape of the problem may have already changed—the lesson is obscured by technological iteration before institutionalization.

Premise Three: The Injured Party Can Coalesce into a Governable Issue. Traditional governance tools also embed a deeper premise: the injured party can identify their own harm and aggregate it into a common issue through the political process. AI platforms simultaneously break this premise at two levels. The individual level: attention fragmentation, judgment degradation, and information filtering bias do not present as explicit harms but are distributed cumulatively across hundreds of millions of interactions—no single individual can clearly identify them. The collective level: even if partial perceptions form, these perceptions, in polarized politics, do not coalesce into a common issue but are instead dismantled into partisan positions.

This is a structural consequence of the social media prism effect: extreme voices are systematically amplified; moderate voices are systematically silenced; the public's perceived degree of opposition far exceeds the actual distribution of opinions (Bail, 2021). The discussion of AI issues in contemporary America clearly exhibits this structure: AI issues are carved up by each party as discursive material; the starting point of discussion is not "what is the harm" but "whose issue is this"—this is the inevitable consequence of party, ideology, religion, and racial identity progressively aligning (sorting) over recent decades: once partisan identity and social identity are bound together, the space for cross-party issue negotiation shrinks, and issue discussion is swallowed by identity affiliation (Mason, 2018). The injured population is first disassembled into mutually estranged partisan collectives; no complete "identifiable injured party" exists for the traditional political process to aggregate. This layer of analysis directly references the sociological literature and is not derived from MCAF.

The simultaneous failure of all three premises means that traditional governance tools can only reach the output readings on the AI black box, while the real operation of power is sealed within parameter matrices. The effective direction is not to increase the number of institutions but to forcibly carve observable interface dimensions onto the black box. The two paths that follow correspond to two mechanisms for this carving. The two paths are derived from the mapping function structure of MCAF (Chapter 4, §2.1), the large/small fiber conversion mechanism of the fiber framework (Chapter 4, §2.4), and the historical comparison of the AI and financial domains (Chapter 34, this chapter §3).

§9.1 The Boundary Audit Principle

The space of behavioral situations inside the black box is extremely vast; exhaustive audit can never catch up. But the boundary of the black box—i.e., the constraint conditions at the time of training and deployment—is registrable and auditable at the design point. The four boundary dimensions identified so far constitute the core of auditing: the optimization dimension list (referencing the conventional and absolute tiers of §7.2); the annotator distribution of the reward model; the prohibited transgression scope of the system prompt; and the macro composition of the training data. Being identified is not being exhausted—the inventory of boundaries must be updated as design forms evolve; the registration obligation is therefore accompanied by an updating obligation. These four dimensions correspond to the decomposition of the visible subspace and the hidden subspace—the goal of auditing is to ensure that hidden subspaces (such as cognitive health, social trust) are not

set as sacrificable costs: per the distinction of §7.1.1—tradeable dimensions must carry non-zero weights with verifiable trade-offs, while non-compensable dimensions may not be discounted at any positive weight (they require hard constraints).

Audit intensity is tiered by risk exposure. For high-risk domains (healthcare, law, financial transactions and investment advisory, education, information distribution and recommendation ranking) with direct impact on life safety, fundamental rights, and social trust, the loss function and system prompt are subject to mandatory ex ante submission for review. For medium-risk domains (hiring, consumer credit, insurance pricing, general-purpose office agents) impacting employment, consumption, and the information environment, the reward model and data composition are submitted for review prior to initial deployment. Tiers within the same industry are determined by the social impact of the specific use case. For low-risk domains, the baseline for transparency is the public disclosure of the optimization dimension list. High and medium-risk domains must simultaneously use at least two independent models for cross-comparison—the two models must come from different training paradigms and different corpus sources; when the output differences exceed a preset threshold, manual review is triggered. Thresholds vary by domain and risk tier; specific parameters are to be formulated and published by a third-party institution independent of both vendors and users, on a per-domain basis.

Boundary auditing cannot cover hidden subspaces that the model spontaneously acquires during training and that are not manifested as boundary-observable features—this is an inherent limitation of the Boundary Audit Principle, not a sign of its technical immaturity.

§9.2 The Mandatory Interface Principle

Boundary auditing only solves observability at the design point. The operation after deployment remains a black box; observation interfaces must be forcibly carved onto the black box to bring key dimensions of the operational state into a similarly verifiable condition. Three types of interfaces:

Encrypted Audit Interface. The platform does not disclose all technical details but only opens controlled access to the design logic of the loss function and key parameters to certified independent audit institutions. The minimum standards for independence include: the audit institution and its affiliates must not hold equity or debt of the audited entity; the proportion of the audit institution's revenue derived from fees paid by the audited entity must be subject to a publicly disclosed upper limit, with the specific proportion to be determined by separate legislation; the attribution of certification authority must be independent of the audited entity and its industry consortia. Specific standards require separate legislation; this section only provides the principled conditions. The auditor signs a legally binding confidentiality agreement. Optional technical implementation directions include encrypted read-only access, differential privacy queries, and zero-knowledge proofs.

Calibration Baseline Interface. The output calibration accuracy of high-risk AI must not be lower than the public baseline—the confidence level annotated by the model must be consistent with the empirical accuracy on an independent test set; those whose deviation exceeds the standard may not be deployed in high-risk scenarios. The institution formulating the public baseline must be independent of model developers and deployers; the baseline dataset must cover representative edge cases in the high-risk domain and be regularly updated—the baseline must be retested and the results publicly disclosed before each major version release. This interface responds to the structural difficulty argued in §8.1—that the RLHF reward model is difficult to audit: it does not require the auditor to reproduce the internal operation of the network, only that the audited party's confidence

declarations be comparable against an external independent test set. In the fiber framework, this corresponds to an institutionalized defense against the "pseudo-small fiber"—using institutions to block the model from masking uncertainty with a tone of certainty (Chapter 4, §2.4).

Data Composition Interface. The macro distribution of the model's training data (language, culture, time period, source type) is disclosed to the public audit institution. What is disclosed is not individual data points but the macro structure—analogous to banks reporting their asset portfolio composition to regulators. If the proportion of AI-generated content in the training data exceeds a threshold, the independent audit institution recommends to the regulatory agency a reduction in deployment scale or the mandatory replenishment of authorized retained native corpora (the "injection" refers to the backflow of retained material, not to conjuring new lived experience); specific execution mechanisms require separate legislation or contractual agreement. This is the institutionalization of preventing the model collapse diagnosed in §1.

The international mutual recognition channel and geopolitical constraints have already been unfolded in §8.2 and will not be repeated here.

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§10 Conclusion: The Limit of the Framework and the Starting Point of Judgment

The financial system took on the order of a century to build the institutional infrastructure capable of ending the cyclical accumulation of shadow debt. Its timeline is given in §3 (252 years, 258 years); the technological capital clock of AI operates on a monthly scale. 252 years is not an option.

In 1955, von Neumann posed the prototype of this question—"Can We Survive Technology?" (von Neumann, 1955). The context at the time was nuclear weapons, but his core insight—that the speed of technological development exceeds the adaptive capacity of human institutions—is even sharper in today's AI context.

But history also tells us something else: no institution begins only when it is ready. The assumption recounted in §3.4 and footnote [3]—that letting the regulated assess their own risks with internal models suffices for safety—was falsified in 2008; the lesson follows from it. Letting developers audit their own models is structurally isomorphic with letting banks assess their own risks.

The disciplinary memory ladder argued in §1—each crisis in the financial domain was followed by the establishment of institutions, but the crisis itself was forgotten within a few generations—implies that the AI domain has yet to experience an equivalent "1907 moment" and cannot afford to wait for it. Waiting for a crisis to teach us is an institutionally unaffordable strategy.

Yet this chapter's argument has also marked its own incompleteness at two key junctures. First: the Public Audit Fund of §5 replaces "outcome-pricing" with "process-pricing," bypassing the two obstacles of untraceability of causal chains and fragmentation of the responsible subject—but the damage discovery mechanism

[3]The 2004 SEC exemption allowing CSE (Consolidated Supervised Entity) large investment banks to use internal models in place of standard net capital rules, the 1996 Basel Market Risk Amendment allowing large banks to calculate market risk capital using internal VaR models, the procyclicality of Basel II and its systemic synchronous failure in 2008, and the structural isomorphism with AI governance, are detailed in Chapter 22, §1 (the 2004 SEC relaxation of net capital rules and the governance clock), §2.3 (MacKenzie's "super-portfolio" pattern: the LTCM strategy homogenization and liquidity collapse, structurally isomorphic with the 2008 CDO market), and §3.1 (the systemic synchronous failure of 2008); the coupling failure of A_{meso} in the Japanese case is addressed in §3.2 of the same chapter.

remains an open gap. An institutional design that acknowledges its own incompleteness is itself an honest starting point. Second: the agent-disorder scenario constructed in §8.5 shows that once an irreversible mismatch occurs between two personal accounts with no one legally responsible, interpersonal damage bypasses the financial compensation loop—the institutional gap on which that scenario rests had not been closed within the period examined in this chapter. But if we wait for it to arrive before starting to build institutions, AI engineering's knowledge half-life on the scale of a few years will very likely wash away that lesson before it can be institutionalized by the next round of technological iteration.

MCAF, as an analytical framework, has reached the limit of its function here. It can diagnose the structure of the problem—why the three premises fail, how the structural defects of the mapping function accumulate cognitive debt, why shadow debt is invisible—but it cannot substitute for political judgment. The ultimate question is not "what is the optimal solution" but "what are we willing to pay for what."

But there is a trap here: framing this choice as a "trade-off between efficiency and subjectivity" already presupposes that "AI can take over cognitive tasks at high quality." This is not the reality. AI output is subject to three constraints—the mean regression of the training distribution, the homogenization bias of RLHF, and hallucinations and confident misjudgments in the long tail. The so-called "let AI take over all cognitive tasks" does not deliver on "efficiency maximization": a vast volume of superficially fluent but fundamentally erroneous outputs is pumped into the system, appearing efficient near the mean but systematically deteriorating in the long tail. Meanwhile, the degradation of human judgment on the other side removes the identifiers of these errors—errors are not digested but sedimented. This process occurs unevenly across different populations, different domains, and different time periods—the very unevenness provides a lifeline for governance windows but also provides openings for narrative arbitrage. The simultaneous deterioration on both sides is not a slide of "efficiency in exchange for subjectivity," but the **simultaneous occurrence of illusory efficiency and real degradation**.

The real judgment, therefore, is not "picking a point between two extremes"—both ends are false propositions. It lies in: in which domains, with what audit conditions, with what revocable mechanisms, is AI allowed to participate in cognitive labor; in which domains must independent human judgment be preserved as the closure point of the loop. This judgment cannot be derived from engineering optimization; it can only be determined through a political process—but that political process must be premised on the Boundary Audit and Mandatory Interfaces of §9, for otherwise, the very observability on which the discussion depends does not exist.

The last thing the MCAF framework can do is to ensure that when this political judgment is made, it is made based on a ledger that is as accurate as possible: known shadow liabilities registered, identified narrative arbitrage flagged, quantified frictional asymmetries entered, "free" restored to its true cost, and the debit-credit asymmetry precisely written on the balance sheet—with the unknown explicitly marked as unknown. The starting point of judgment is not within this book—judgment begins after this, premised on a clean ledger.

The two paths proposed in §9—Boundary Audit and Mandatory Interfaces—are not institutional blueprints but two mechanisms for forcibly carving observable interface dimensions onto the black box. The jump from these two paths to specific parameters requires a political process and pilot iterations.

This is not a solution, but a beginning—just as the 8% capital adequacy ratio of Basel I in 1988 was not a perfect solution, but a beginning. AI governance needs the counterpart of this principle: the key parameters of AI systems cannot be unilaterally set by developers, still less can developers, driven by commercial competition, accelerate the diffuse global impact of key parameters; they must be subject to public audit. Once the principle

is established, specific standards can be iterated.

In the language of the Clock Room: these clocks are not merely displaying readings—they are looking at the time on behalf of billions of people. Letting the clockmaker calibrate their own clock alone is the failed lesson of Basel II in the domain of financial governance. This book hopes the AI domain will not repeat it.

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